

Republic of Arab Egypt
Ministry of Health and Population
Central Administration for Human Resources Development
General Administration of Technical Health Education
Education Department

SECONDARY TECHNICAL NURSING SCHOOL: ANATOMY AND PHYSIOLOGY FOR FIRST GRADE

Prepared and Reviewed by

Prof. Dr. / Mandour Mahmoud Abdel Rahman
Professor of Anatomy
Kasr Al-Ainy Faculty of Medicine - Cairo University

Prof. Dr. / Sanaa Mohamed Morsi
Professor of Physiology
Kasr Al-Ainy Faculty of Medicine - Cairo University

Fundamentals of Anatomy

Chapter One

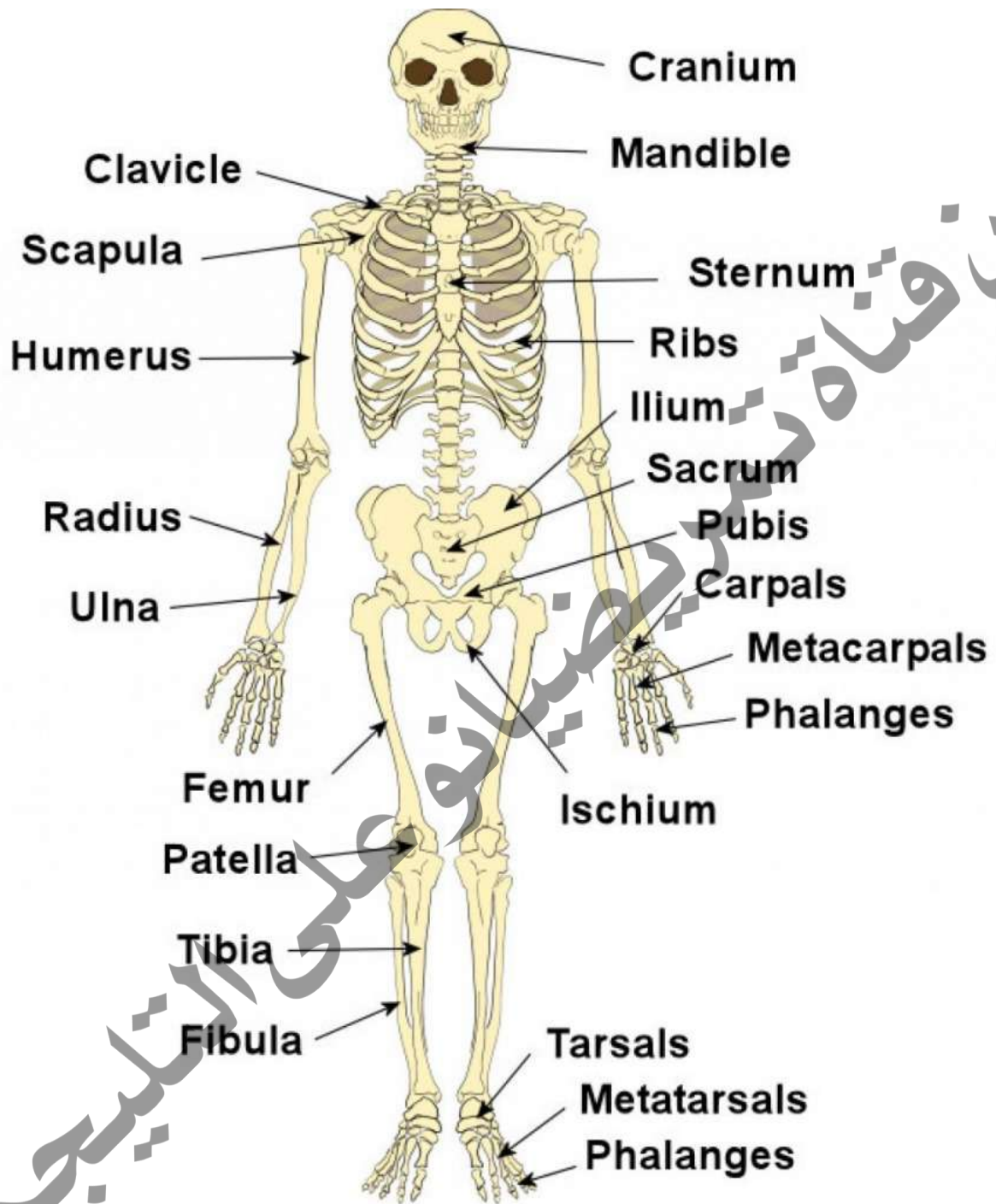
The Bones

Types of Bones:

1. **Long Bones:**
Elongated bones with two ends and a shaft between them. The shaft has a cavity containing bone marrow.
Example: Humerus and Femur bones.
2. **Short Bones:**
They are either round, cubical, or pyramidal in shape.
Example: Carpal bones of the hand.
3. **Irregular Bones:**
They are short bones but do not have a distinct shape.
Example: Vertebrae.
4. **Flat Bones:**
They are thin and flat bones.
Example: Skull bones and Scapula.
5. **Pneumatic Bones:**
They are bones in the skull that contain air-filled spaces in some parts called paranasal sinuses. They are important for voice resonance as they are connected to the nasal cavity.
6. **Sesamoid Bones:**
They are small bones found inside the tendons of some muscles.
Example: The Patella located in front of the knee.

Parts of the Skeleton:

1. **Axial Skeleton:** Located in the middle of the body and includes:
 - The Skull.
 - The Vertebral Column.
 - The Ribs.
 - The Sternum.
2. **Appendicular Skeleton:** Located on the sides of the body and includes:
 - Bones of the Upper Limb.
 - Bones of the Lower Limb.



The Skull

It consists of 21 bones separated from each other by irregular sutures. Some bones are single (in the midline) like: Frontal and Occipital. Some are paired (on the sides) like Parietal and Temporal.

The mandible connects with the skull at a joint located directly in front of the ear opening.

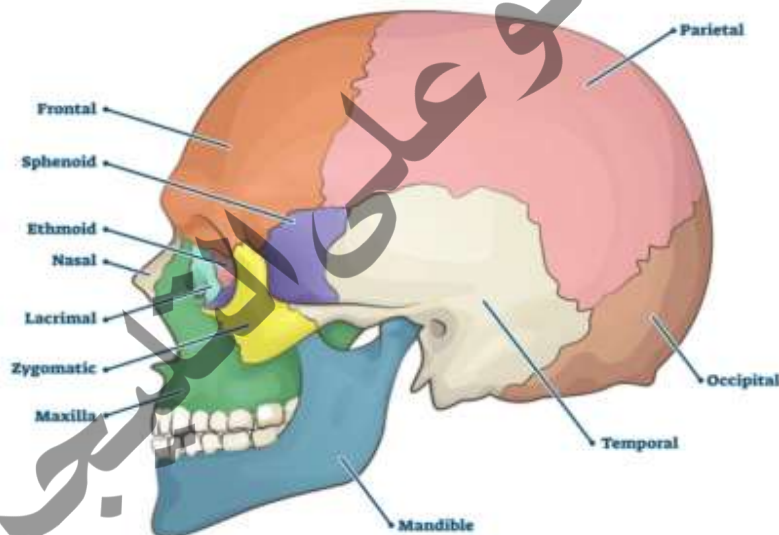
The skull consists of 3 parts:

- Vault (Superior part).
- Face (Anterior part).
- Base (Inferior part).

The base of the skull contains several foramina through which nerves and blood vessels connected to the brain pass. Therefore, a fracture of the base of the skull is more dangerous because it may lead to bleeding inside the skull due to injury to these blood vessels. This may lead to bleeding from the nose and ears.

Adult Skull

HUMAN SKULL



The Skull of a Newborn Infant

Characterized by the following:

1. It has no sinuses.
2. It has no air sinuses.
3. The face is small.
4. It has several fontanelles.

Fontanelles:

- These are areas in the skull of a newborn infant where the bones have not yet formed, but contain membranes that are replaced with bones as the child grows.
- The number of fontanelles is six, but the most important are the anterior and posterior fontanelles.

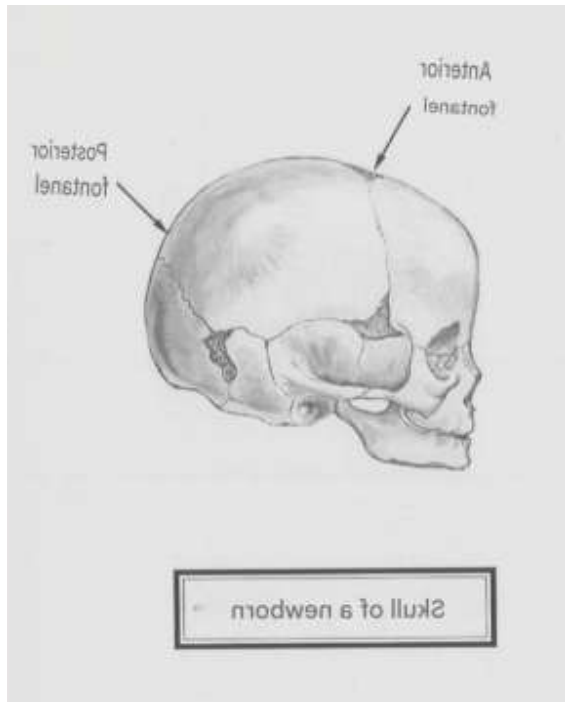
The following table compares the anterior and posterior fontanelles:

Aspect of Comparison	Anterior Fontanelle	Posterior Fontanelle
Size	Large	Small
Length	3 cm	1 cm
Shape	Diamond-shaped	Triangular
Closure Time	After a year and a half (or two years)	Within three months

Importance of Fontanelles:

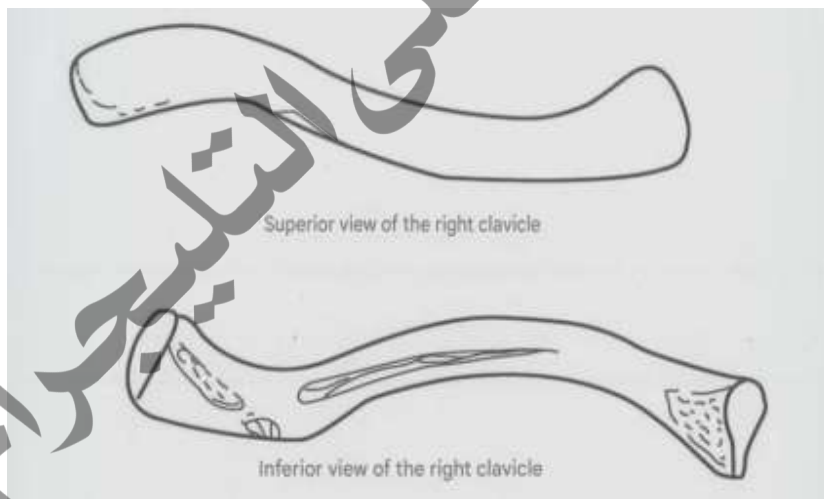
1. Determining the child's age.
2. Determining the fetal position at birth.
3. Diagnosing rickets in children (if their closure is delayed).
4. Diagnosing dehydration in children (if they are sunken).
5. Diagnosing increased cerebrospinal fluid pressure (if they are bulging).
6. Facilitating birth because they allow the skull to contract.
7. May be used in emergencies for intravenous injection, as there is a large venous sinus under the anterior fontanelle where injection can be done if no other vein is found.

Skull of a Newborn Infant



Bones of the upper limb:

Clavicle:



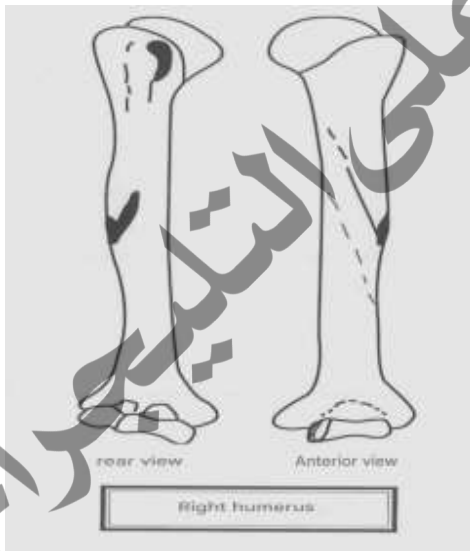
Located in the pectoral girdle at the front. It extends between the sternum and scapula bones.

Scapula:

Located in the pectoral girdle at the back. It connects with the clavicle and humerus bones.



Humerus Bone:



Extends between the shoulder and elbow joints. It has a head that articulates with the scapula at the shoulder joint. It articulates inferiorly with the ulna and radius bones at the elbow joint.

Radius Bone:

Located in the forearm on the lateral side.

It participates in four joints:

1. Elbow.
2. Wrist.
3. Superior Radioulnar.
4. Inferior Radioulnar.



Anterior view

posterior view

Ulna Bone:

Located in the forearm region on the medial side.

It participates in three joints:

1. Elbow.
2. Superior Radioulnar.
3. Inferior Radioulnar.

It does not participate in the wrist joint.

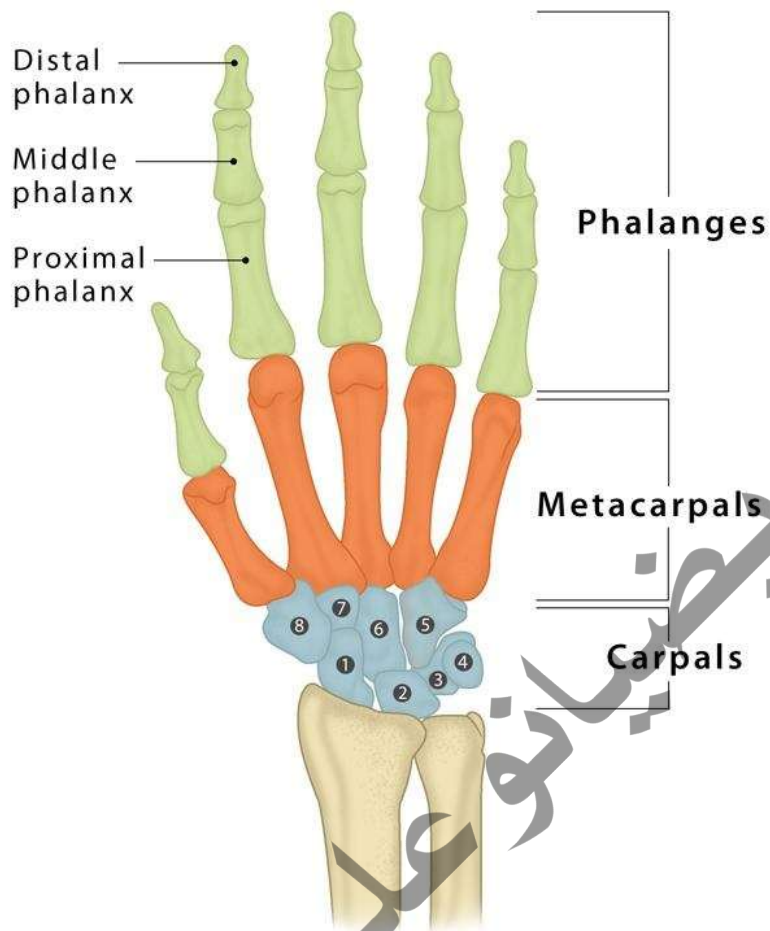


Bones of the Hand:

The hand is divided into three regions: Carpus - Metacarpus - Fingers.

- **Carpus:** 8 carpal bones arranged in two rows.
- **Metacarpus:** 5 metacarpal bones.
- **Fingers:** 14 phalanges: "In each finger except the thumb which has only 2."

Hand Bones



- **Main Joints of the Upper Limb:**

1. Shoulder joint: Between scapula and humerus.
 2. Elbow joint: Between humerus and forearm.
 3. Wrist joint: Between forearm and hand.
-

Table summarizing bones of the upper limb

Region	Bone(s)	Position
Pectoral Girdle	Clavicle Scapula	At the front At the back
Arm	Humerus	Between shoulder and elbow joints
Forearm	Ulna Radius	On the medial side On the lateral side
Hand	8 Carpals (in two rows) 5 Metacarpals 14 Phalanges (2 in thumb, 3 in others)	Wrist Palm Fingers

Bones of the Lower Limb

Pelvic Bone:

The two pelvic bones connect anteriorly at the pubic symphysis joint. Posteriorly, the sacrum connects between the two pelvic bones.

The pelvic bone connects externally with the femur bone at the hip joint.

The pelvic bone consists of three parts:

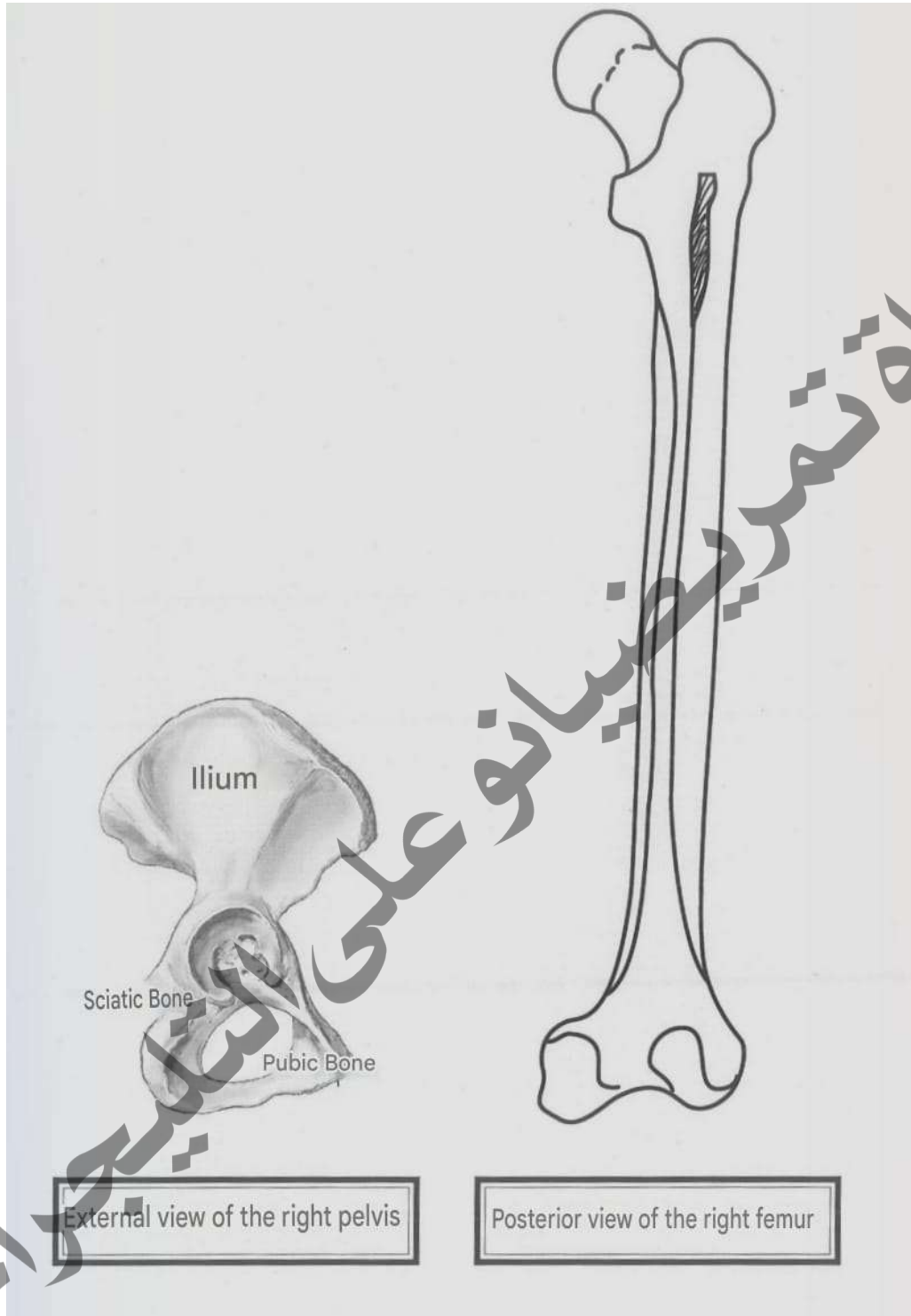
1. Pubis bone (anteriorly).
2. Ischium bone (posteriorly).
3. Ilium bone (superiorly).

These three parts are separate from each other at birth, connected by cartilage, but they fuse together at puberty.

Femur Bone:

Connects superiorly with the pelvic bone at the hip joint and inferiorly with the tibia at the knee joint.

It is the longest and strongest bone in the body.

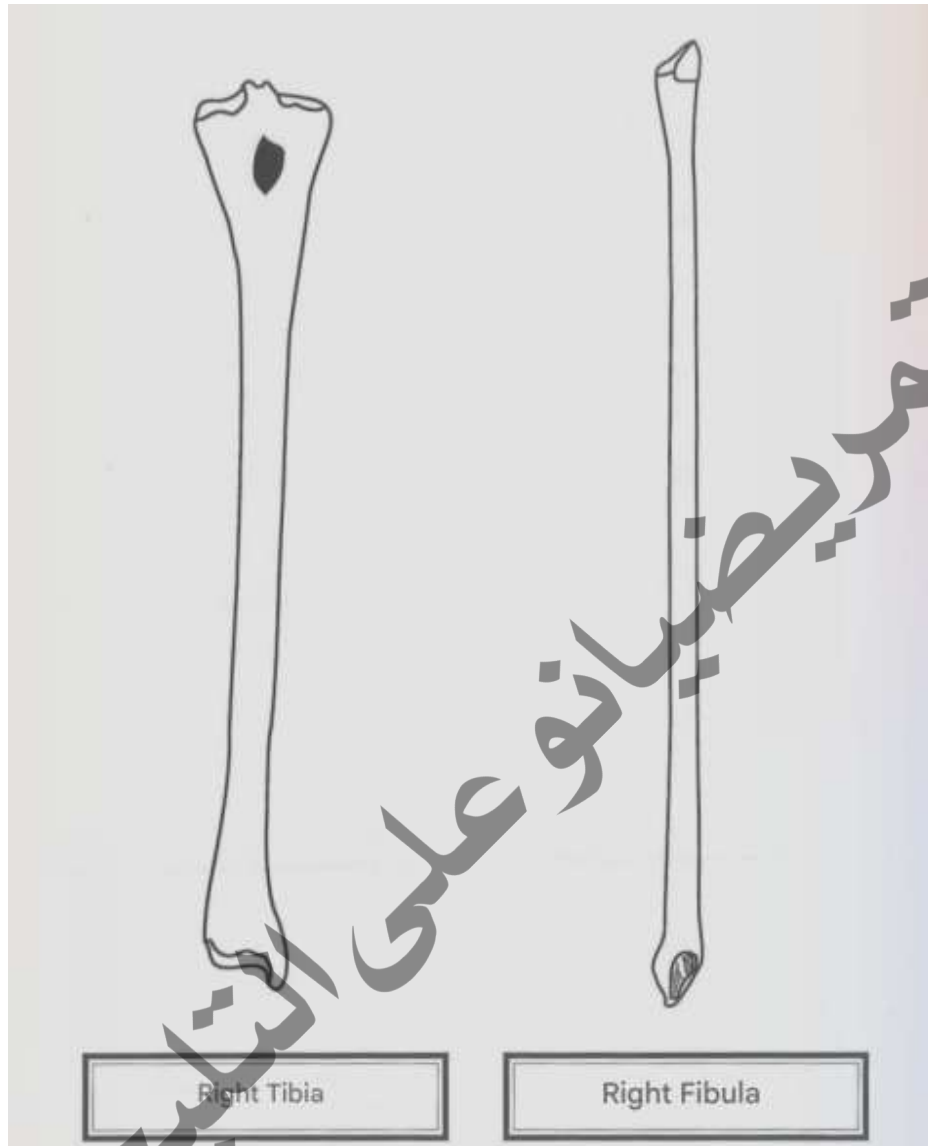


Tibia:

Located in the leg on the medial side. It participates in bearing the body's weight.

Fibula:

Located in the leg on the lateral side. It does not participate in bearing body weight.



Bones of the Foot:

The foot is divided into three regions: Tarsus – Metatarsus – Toes.

- **Tarsus:** 7 tarsal bones.

- **Metatarsus:** 5 metatarsal bones.
- **Toes:** 14 phalanges.

The bones of the foot are arched longitudinally and transversely to protect the nerves and blood vessels in the sole of the foot. Some people suffer from flat feet, which causes them severe pain when standing due to pressure on the nerves and blood vessels in the sole of the foot.

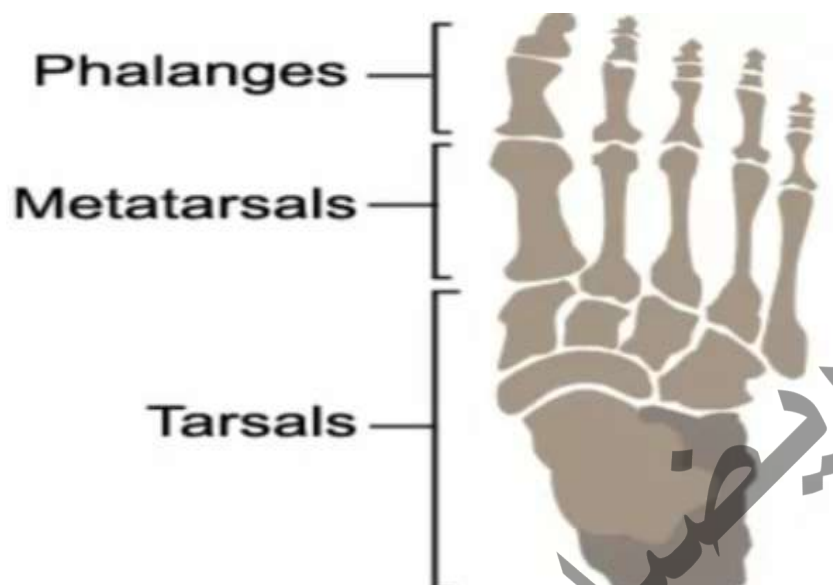


Table summarizing bones of the lower limb

Region	Bone(s)	Position
Pelvic Girdle	Pelvic bone	On both sides of the pelvis
Thigh	Femur	Between hip and knee joints
Leg	Tibia Fibula	On the medial side On the lateral side
Foot	7 Tarsals 5 Metatarsals 14 Phalanges (2 in big toe)	Ankle Sole of foot Toes

Joints of the Lower Limb:

1. Hip joint: Between pelvis and femur.
2. Knee joint: Between femur and leg.
3. Ankle joint: Between leg and foot.

Bones of the Thoracic Cage

(1) Sternum Bone:

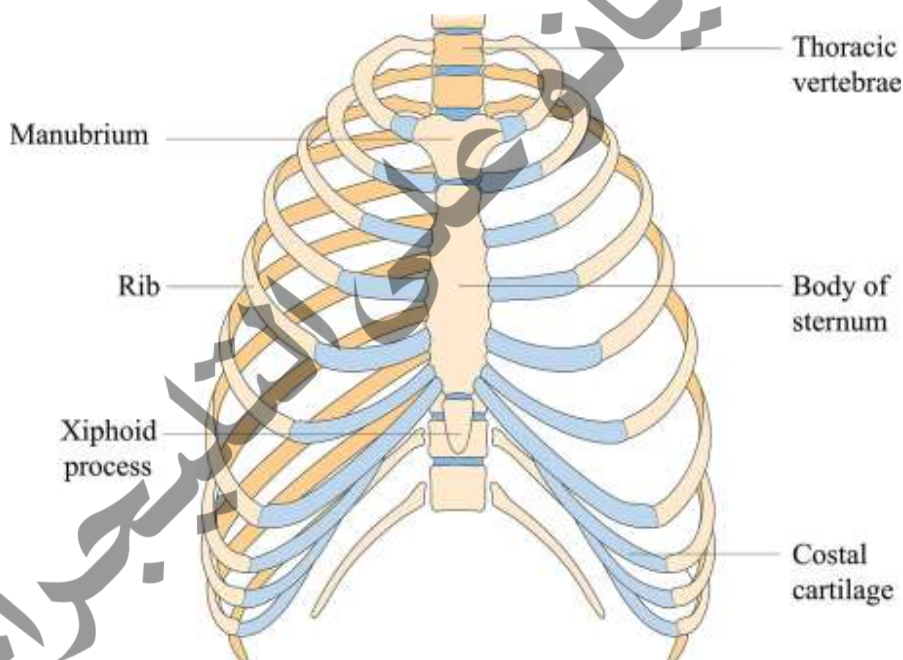
Located at the front and consists of 3 parts: Manubrium - Body - Xiphoid process. The manubrium and body meet at the sternal angle, where the second rib attaches.

(2) **Thoracic Vertebrae:** At the back.

(3) **Ribs:** On each side.

Types of Ribs:

1. **True Ribs:** Those that connect directly to the sternum anteriorly, represented by pairs 1-7.
2. **False Ribs:** Those that do not connect directly to the sternum anteriorly, represented by pairs 8-10.
3. **Floating Ribs:** Those that do not connect to anything anteriorly, represented by pairs 11-12.



The Vertebral Column

It consists of 33 vertebrae extending behind the trunk, distributed as follows:

- **7 Cervical vertebrae:** In the neck region.
- **12 Thoracic vertebrae:** In the chest region.
- **5 Lumbar vertebrae:** In the abdominal region.
- **Sacral vertebrae (one piece):** In the pelvic region between the pelvic bones.
- **Coccygeal vertebrae (one piece):** Also in the pelvic region, below the sacrum.

Parts of a Vertebra:

Each vertebra contains a large foramen through which the spinal cord passes.

- **Body of the vertebra:** Located in front of the foramen.
- **Spinous process:** Located behind the foramen.
- **Transverse processes:** On the sides of the foramen.

Curves of the Vertebral Column:

At birth, the vertebral column is shaped like a single curve convex backward due to the fetal position inside the uterus.

When the child can raise his head, the cervical vertebrae become concave and convex forward.

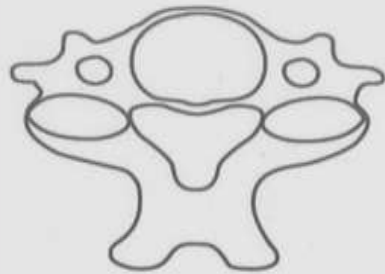
When the child can stand, the lumbar vertebrae become concave and convex forward.

While the rest of the regions remain convex backward, thus the vertebral column has 4 curves:

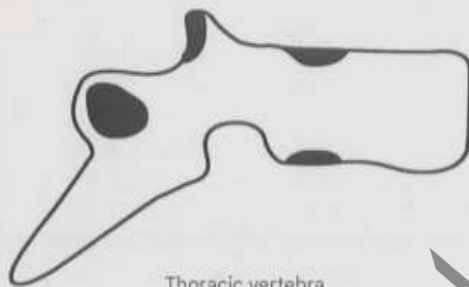
- In the neck region: Curve convex forward (Lordosis).
- In the chest region: Curve convex backward (Kyphosis).
- In the abdominal region: Curve convex forward (Lordosis).
- In the pelvic region: Curve convex backward (Kyphosis).

Functions of the Vertebral Column:

- Protect the spinal cord.
- Bear the body's weight.
- Form the central axis of the body.



Cervical vertebra

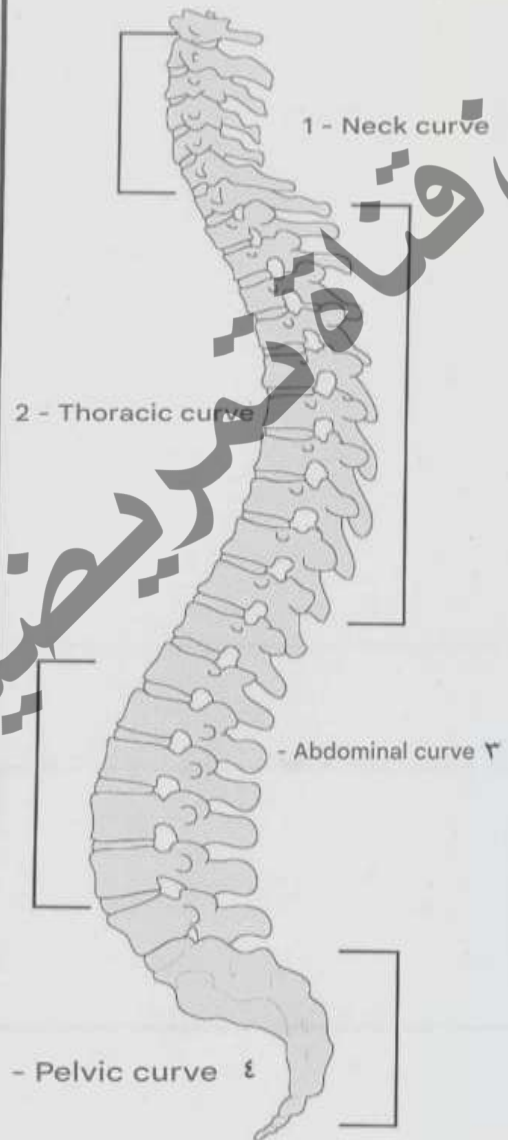


Thoracic vertebra



Lumbar vertebra

Lumbar vertebra



Left view of the spine

The Pelvis

The pelvis consists of:

Bones:

1. Right pelvic bone (on the right side).
2. Left pelvic bone (on the left side).
3. Sacrum (posteriorly).
4. Coccyx (below the sacrum).

Joints:

1. Right sacroiliac joint (between right pelvic bone and sacrum).
2. Left sacroiliac joint (between left pelvic bone and sacrum).
3. Sacrococcygeal joint (between sacrum and coccyx).
4. Pubic symphysis (between the two pelvic bones anteriorly).

The pelvis consists of two parts:

1. **False Pelvis:** The superior, wide part.
2. **True Pelvis:** The inferior, narrow part, which is important during childbirth.

Main differences between male and female pelvis:

Aspect of Comparison	Female Pelvis	Male Pelvis
Inlet	Wide and kidney-shaped	Heart-shaped
Outlet	Wide	Narrow
Cavity	Wide and shallow (not deep)	Narrow and deep
Pubic Arch	Wide (more than 90 degrees)	Narrow (less than 90 degrees)
Sacrum	Wide and short	Narrow and long
Coccyx	Movable	Immovable

REVIEW QUESTIONS CHAPTER 1

1. Answer the following questions:

- Mention the parts of the axial skeleton.
- The skull consists of 3 parts, what are they and where are they located?
- Compare the anterior and posterior fontanelles, mentioning three of their functions.
- Mention the parts of the pectoral girdle.
- Mention the main joints of the upper limb.
- Mention the types of ribs.

2. Complete the following sentences:

- The pelvis consists of two parts, which are and
- At birth, the vertebral column is shaped like convex backward. When the child can raise his head, the vertebrae become concave and convex towards, and when the child can stand, the vertebrae become concave and convex towards
- From the bones of the lower limb, the does not participate in bearing body weight.

3. Choose the correct answer from the following:

a) Examples of flat bones:

- () Patella.
- () Scapula.
- () Vertebrae.
- () Phalanges.

b) Characteristics of the male pelvis:

- () Inlet is kidney-shaped.
- () Outlet is wide.
- () Cavity is deep.
- () Cavity is wide.

CHAPTER TWO

The Nervous System

The nervous system consists of:

1. **Central Nervous System (CNS):** Includes the brain and spinal cord.
2. **Peripheral Nervous System (PNS):** Includes:
 - a. Cranial nerves (12 pairs).
 - b. Spinal nerves (31 pairs).

The Brain

Located inside the skull. Consists of:

1. Cerebrum (largest part).
2. Cerebellum.
3. Brainstem.

Cerebrum:

Divided into right and left hemispheres, each specializing in the opposite side of the body. Each hemisphere consists of four lobes.

The following table shows the main function and location of each lobe:

Lobe	Location	Function
Frontal	In the front	Contains motor centers
Parietal	On top	Contains sensory centers
Temporal	At the bottom	Contains hearing and memory centers
Occipital	In the back	Contains vision centers

Note:

The speech center in most people (who are right-handed) is located in the left hemisphere of the cerebrum. Therefore, if a stroke or hemorrhage occurs in the left side of the brain, the patient suffers from hemiplegia and loss of sensation on the right side of the body, accompanied by loss of the ability to speak. If the stroke or hemorrhage is in the right side of the brain, the patient only suffers from hemiplegia and loss of sensation on the left side of the body.

Cerebellum:

Located below the cerebrum at the back. Consists of right and left hemispheres. It has two functions:

1. Maintaining body balance.
2. Controlling voluntary movements.

Brainstem:

Consists of:

1. **Midbrain:** Contains some visual and auditory centers.
2. **Pons:** It is the connection between the cerebrum and cerebellum.
3. **Medulla Oblongata:** It is the most important part of the brainstem because it contains the vital centers controlling the heart, respiration, and blood pressure.

The medulla oblongata exits the skull and its extension is called the spinal cord, which passes inside the vertebral column.

Spinal Cord:

Length is 45 cm. It is located inside the vertebral column.

The spinal cord begins below the skull and ends at the level of the first lumbar vertebra (L1) of the vertebral column. Therefore, the vertebral column below the first lumbar vertebra does not contain the spinal cord but contains a bundle of spinal nerves that supply the lower half of the body, called the *Cauda Equina* (horse's tail). This is what is anesthetized during spinal anesthesia by injecting the anesthetic between the fourth and fifth lumbar vertebrae (L4-L5).

The brain and spinal cord are surrounded by three membranes (Meninges):

- **Dura Mater:** The outer layer, a thick layer that lines the skull.
- **Arachnoid Mater:** The middle layer.
- **Pia Mater:** The inner layer that lines the brain

Located between the arachnoid mater and the pia mater is the cerebrospinal fluid, which protects the brain and spinal cord from impacts and high environmental temperatures.

The Peripheral Nervous System

Spinal Nerves:

- There are 31 pairs that exit the spinal cord through foramina between the vertebrae of the vertebral column on both sides.
- They are responsible for movement and sensation of the entire body except the head.

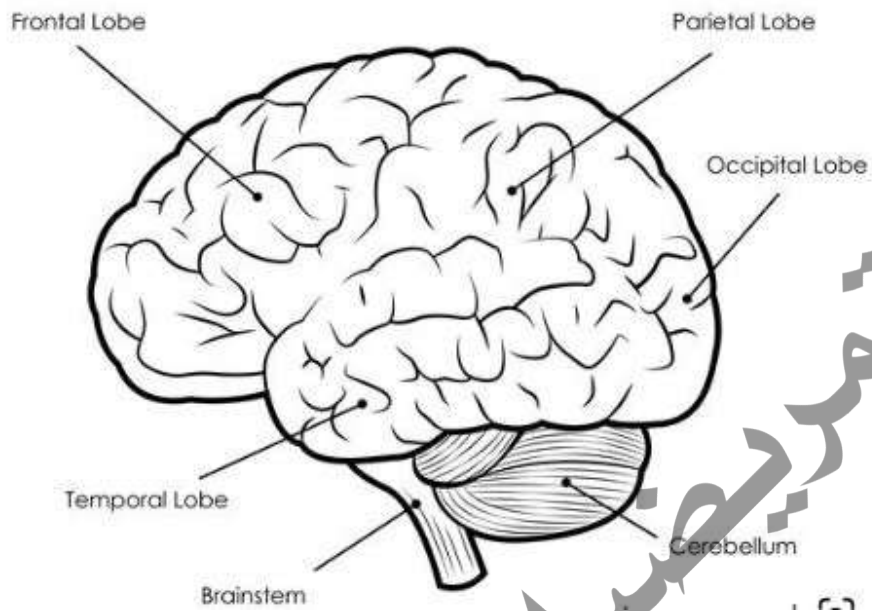
Cranial Nerves:

There are 12 pairs that exit the brain through foramina in the base of the skull.

The following table shows the name and function of each:

#	Name	Function
1	Olfactory	Sense of smell
2	Optic	Sense of vision
3	Oculomotor	Innervates all eye muscles except two
4	Trochlear	Innervates one eye muscle
5	Trigeminal	Branches into 3 nerves:
	<i>Ophthalmic:</i>	Sensation of the eye and forehead.
	<i>Maxillary:</i>	Sensation of the upper jaw.
	<i>Mandibular:</i>	Sensation of the lower jaw + Innervation of chewing muscles.
6	Abducens	Innervates one eye muscle
7	Facial	Innervates facial muscles + 2 salivary glands.
8	Vestibulocochlear	Innervates the inner ear; functions are hearing and balance.
9	Glossopharyngeal	Sensation and taste from the tongue + Parotid gland.
10	Vagus	Parasympathetic nervous system.
11	Accessory	Innervates neck and larynx muscles.
12	Hypoglossal	Innervates tongue muscles.

Parts of the Brain



REVIEW QUESTIONS CHAPTER 2

1. Answer the following questions:

- Mention the location and function of the lobes of the cerebrum.
- Mention the parts of the brainstem and which is the most important part and why?
- The brain and spinal cord are surrounded by three membranes (meninges), what are they?

2. Complete the following sentences:

- The cerebellum is located and has two functions: and
- Motor centers are located in the lobe, which is located
- The two nerves that innervate the salivary glands are and

3. Choose the correct answer from the following:

a) The nerve that innervates the chewing muscles is called:

- ☐ Abducens nerve.
- ☐ Glossopharyngeal nerve.
- ☐ Maxillary nerve.
- ☐ Mandibular nerve.

b) The nerve responsible for the parasympathetic nervous system is:

- ☐ Vagus nerve.
- ☐ Trochlear nerve.
- ☐ Trigeminal nerve.
- ☐ Accessory nerve.

c) Hearing centers are located in:

- ☐ The frontal lobe of the cerebrum.
 - ☐ The parietal lobe of the cerebrum.
 - ☐ The temporal lobe of the cerebrum.
 - ☐ The medulla oblongata.
-

CHAPTER THREE

The Circulatory System

Includes the heart and blood vessels.

The Heart

A conical muscular organ with an apex and a base.

Base:

On the right side, formed by the atria.

Apex of the heart:

Located on the left side of the chest, between the 5th and 6th ribs, 9 cm from the midline. The mitral valve sound can be heard clearly over it.

The heart has 4 chambers and 4 valves.

The four heart valves:

Valve	Location	Blood Type
Mitral (Bicuspid)	Between left atrium & left ventricle	Oxygenated
Tricuspid	Between right atrium & right ventricle	Deoxygenated
Aortic	Between left ventricle & aorta	Oxygenated
Pulmonary	Between right ventricle & pulmonary artery	Deoxygenated

Arteries supplying the heart:

The right and left coronary arteries supply the heart with O₂.

While waste blood with CO₂ exits via the coronary venous sinus to the right atrium.

Note:

In case of narrowing of a coronary artery or one of its branches due to a clot forming inside it, the oxygen supplied to the heart is insufficient, especially during exertion, causing the patient to feel pain, a condition called angina pectoris. The artery may become completely blocked, leading to the death and fibrosis of a part of the heart muscle.

The four chambers of the heart:

Chamber Name	Openings	Blood Type
Right Atrium	Superior vena cava Inferior vena cava Coronary venous sinus Opening of tricuspid valve	Deoxygenated
Right Ventricle	Pulmonary trunk Opening of tricuspid valve	Deoxygenated
Left Atrium	Pulmonary veins Opening of mitral valve	Oxygenated
Left Ventricle	Aorta Opening of mitral valve	Oxygenated

Circulatory Pathways

1. **Systemic Circulation (Greater):**

The left ventricle pumps oxygenated blood to the aorta, which distributes it to all parts of the body. After oxygen is consumed, deoxygenated blood returns to the right atrium via the superior and inferior vena cava.

2. **Pulmonary Circulation (Lesser):**

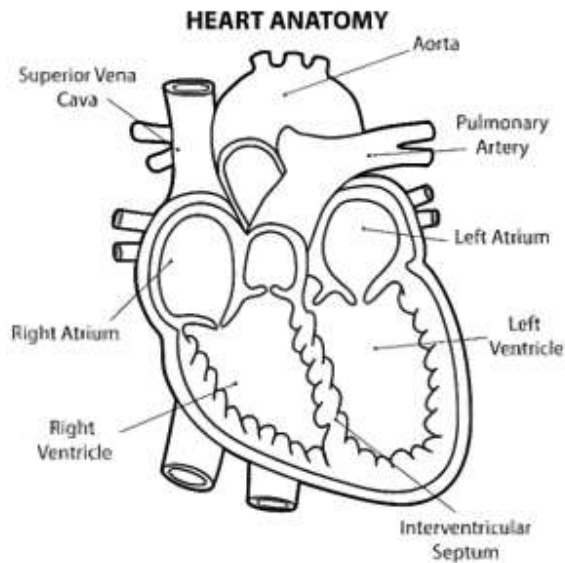
The right ventricle pumps deoxygenated blood to the pulmonary trunk, which carries it to the lungs where gas exchange occurs. Then, oxygenated blood returns to the left atrium via pulmonary veins.

3. **Portal Circulation:**

Veins from the digestive system (from the lower esophagus to the anal canal) join with the splenic vein to form the portal vein, which empties into the liver where nutritional metabolism and detoxification occur.

In case of liver cirrhosis, blood is diverted to other pathways, causing esophageal varices and hemorrhoids due to the engorgement of veins in these areas from the diverted blood.

The Heart



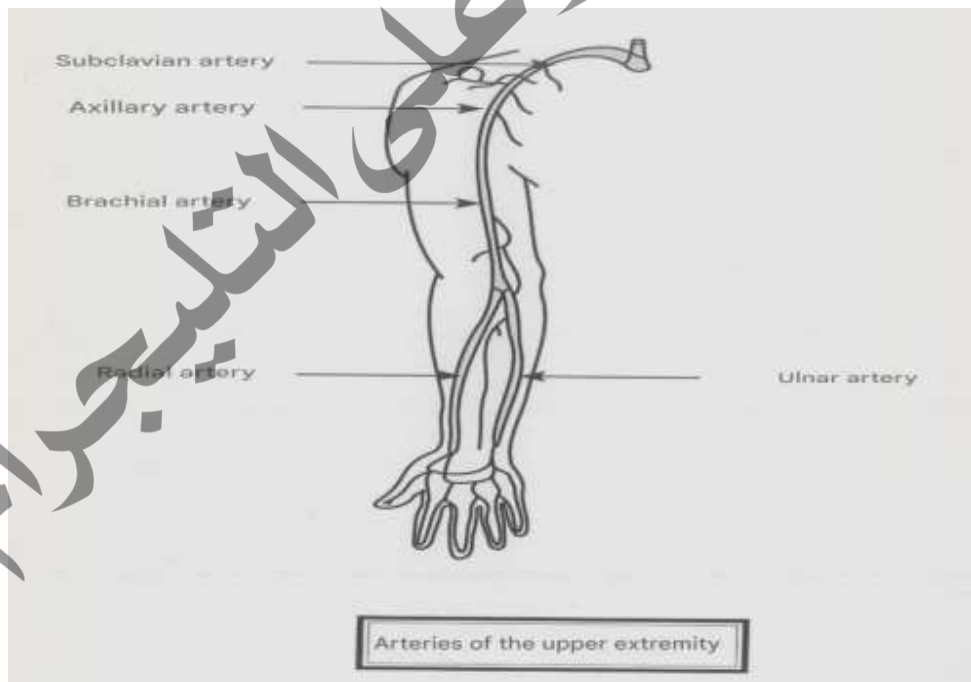
Names and locations of upper limb arteries:

Artery Name	Location
Axillary artery	Axilla (armpit)
Brachial artery	Arm, medial side (used for measuring blood pressure)
Radial artery	Lateral side of forearm (used for measuring pulse)
Ulnar artery	Medial side of forearm
Palmar arterial arches	Palm of the hand

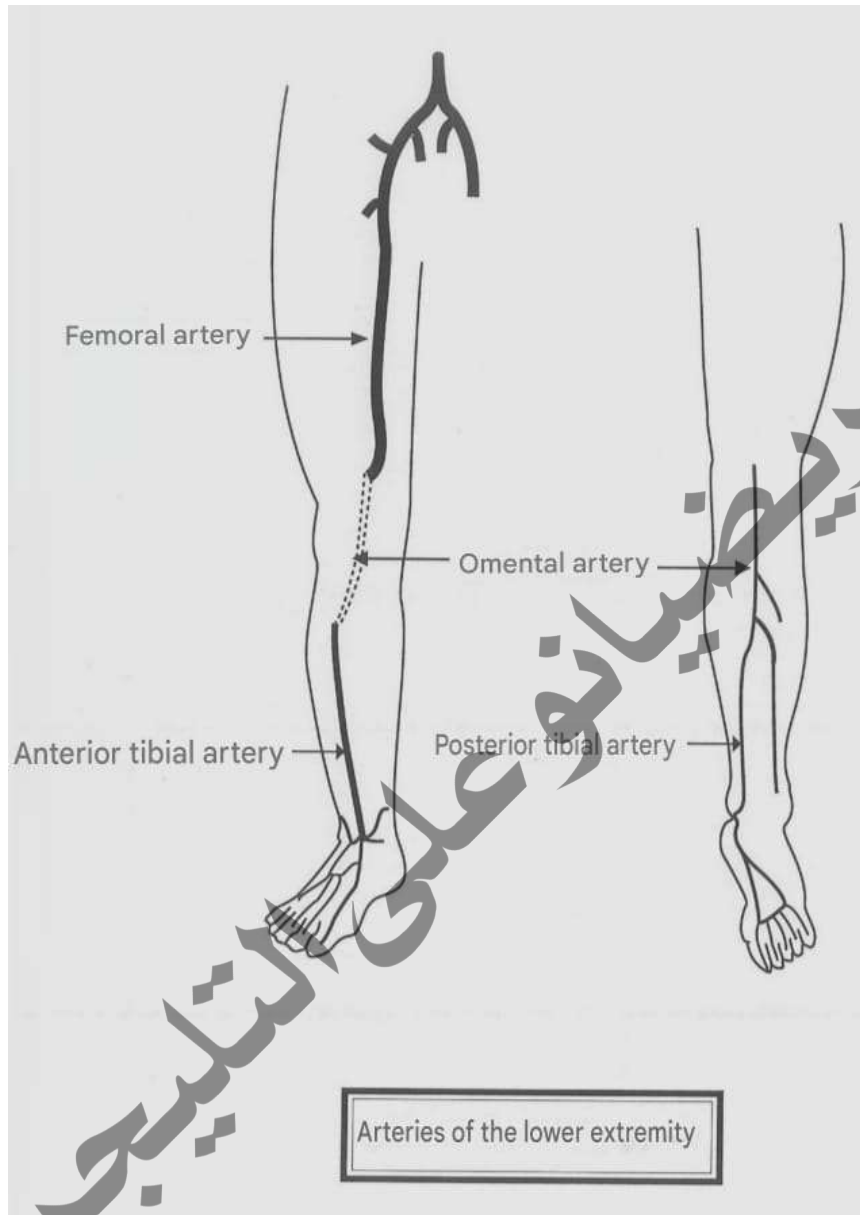
Names and locations of lower limb arteries:

Artery Name	Location
Femoral artery	In the thigh, anteriorly
Popliteal artery	Behind the knee, in the popliteal fossa
Anterior tibial artery	In the leg, anteriorly
Posterior tibial artery	In the leg, posteriorly
Dorsalis pedis artery	Dorsum of the foot (pulse is weak but sensing it is important in diabetic patients)
Plantar arteries	Sole of the foot

Arteries of the Upper Limb



Arteries of the Lower Limb



Names and locations of upper limb veins:

Vein Name	Location
Subclavian vein	Under the clavicle
Axillary vein	Axilla
Cephalic vein	Lateral side of the upper limb
Basilic vein	Medial side of the upper limb
Median cubital vein	In front of the elbow (commonly used for IV injection)

Names and locations of lower limb veins:

Vein Name	Location
Femoral vein	In the thigh, anteriorly
Popliteal vein	Behind the knee, in the popliteal fossa
Long saphenous vein	On the medial side of the thigh and leg (used for IV injection)
Short saphenous vein	In the leg, posteriorly

The heart is surrounded by two spaces:

1. **Fibrous Pericardium:**

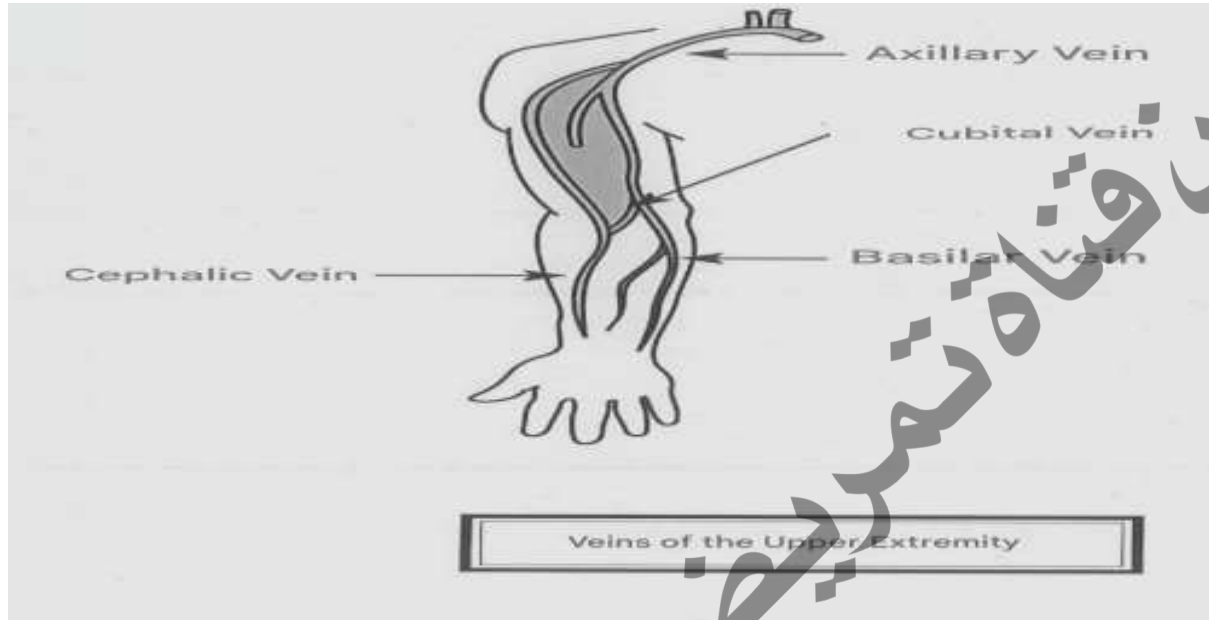
A strong membrane that protects the heart, holds it in place, and prevents it from over-expanding.

2. **Serous Pericardium:**

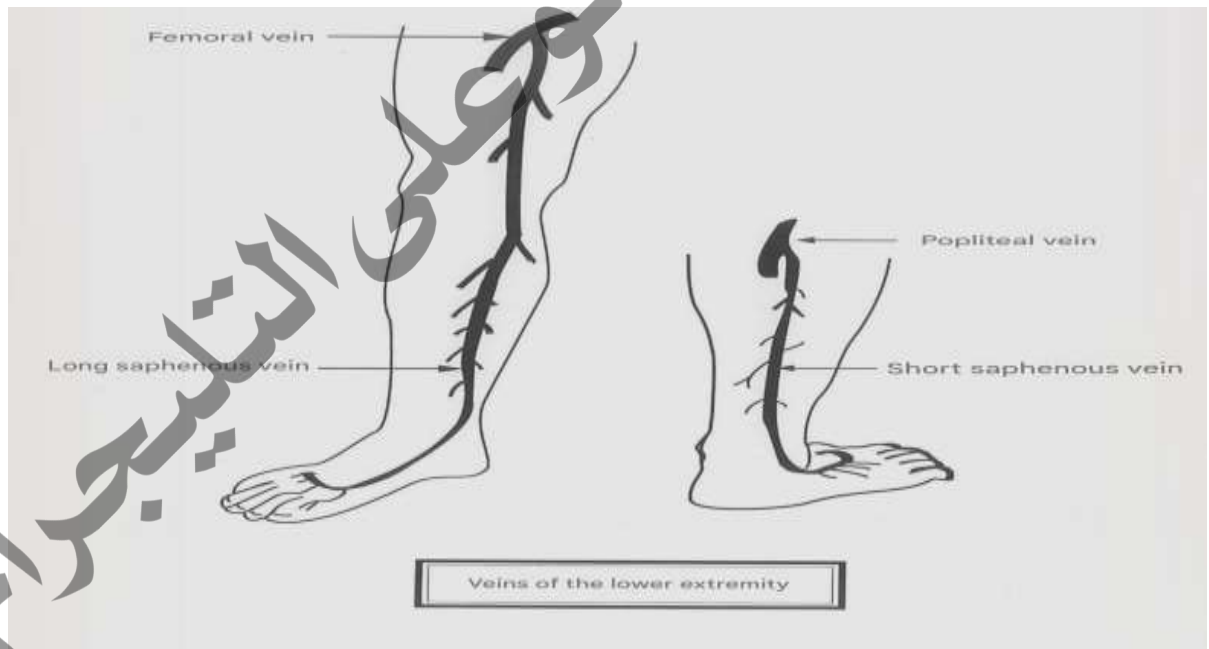
A thin membrane lining the heart, consisting of two layers with a narrow space between them. It facilitates heart movement by preventing friction.

من فضاء نهريضانو على التليجرام

Veins of the Upper Limb



Veins of the Lower Limb



REVIEW QUESTIONS CHAPTER 3

1. **Answer the following questions:**

- Where is the apex of the heart located and what is its importance?
- Where are the four heart valves located and what type of blood is in each valve?
- Mention what you know about the portal circulation.
- Name the arteries of the upper limb and their locations.
- Name the veins of the lower limb and their locations.
- The heart is surrounded by membranes. What are they and what is their importance?

2. **Complete the following sentences:**

The artery from which the pulse is usually taken is, the one from which pressure is measured is, while the most commonly used vein for injection is

The arteries that supply the heart with oxygenated blood are, while carries deoxygenated blood from the heart to

3. **Choose the correct answer from the following:**

The mitral valve is located between:

- () Left ventricle and aorta
- () Left atrium and right ventricle
- () Left atrium and left ventricle
- () Right atrium and right ventricle

The portal vein empties into:

- () Inferior vena cava
 - () Superior vena cava
 - () Right atrium
 - () Liver
-

CHAPTER FOUR

The Lymphatic System

Includes lymphatic vessels and lymph nodes.

Lymphatic Vessels:

Begin in the tissues where lymphatic capillaries converge to collect excess high-density tissue fluid (which cannot enter blood capillaries) and carry it to the lymph node where this fluid is filtered. Another lymphatic vessel exits the lymph node to empty into another lymph node. This process is repeated several times until the lymph fluid is purified, and finally, the last lymphatic vessel empties into the nearest vein.

Lymph Nodes:

1. Filter lymph fluid from impurities (bacteria, dust, dead cells, cancer cells).
2. Produce lymphocytes.
3. Produce antibodies.

Lymph Node Groups:

Lymph nodes are grouped in specific locations. Each group is responsible for defending specific areas of the body. If an injury occurs in a specific area, the responsible lymph nodes swell until the affected area heals.

The following table shows the main lymph node groups and the body parts each group defends:

Group	Location	Drained Areas
Cervical nodes	Under sternocleidomastoid muscle	Head and neck
Axillary nodes	Axillary fossa	Upper limb and breast
Thoracic nodes	Mediastinum	Heart and lungs
Aortic nodes	On both sides of the aorta	Abdominal viscera
Iliac nodes	In the pelvis	Pelvic viscera
Inguinal nodes	Inguinal fossa	Lower limb and external genital organs

The Spleen

A fist-sized, wedge-shaped organ. Similar in structure to a lymph node.

Location: In the upper left part of the abdomen, behind the stomach. Under ribs 9, 10, and 11.

It has two surfaces:

- *External surface:* Smooth - contacts the diaphragm.
- *Internal surface:* Irregular - contacts the stomach, colon, and left kidney.

Functions of the Spleen:

1. Stores blood and can contract in case of hemorrhage to compensate for some lost blood.
2. Destroys worn-out red blood cells and sends the hemoglobin inside them to the liver to form bile.
3. Produces white blood cells.

The Thymus Gland

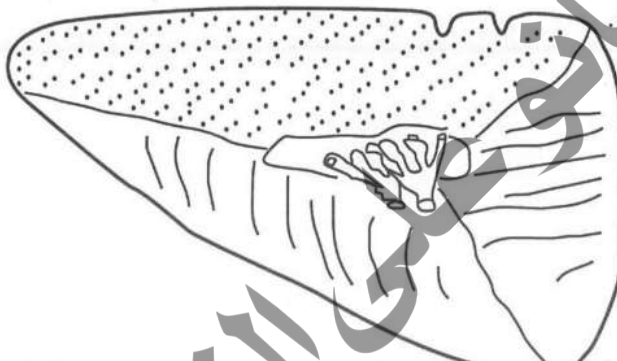
An endocrine gland that belongs to the lymphatic system.

Located in front of the heart, between it and the sternum.

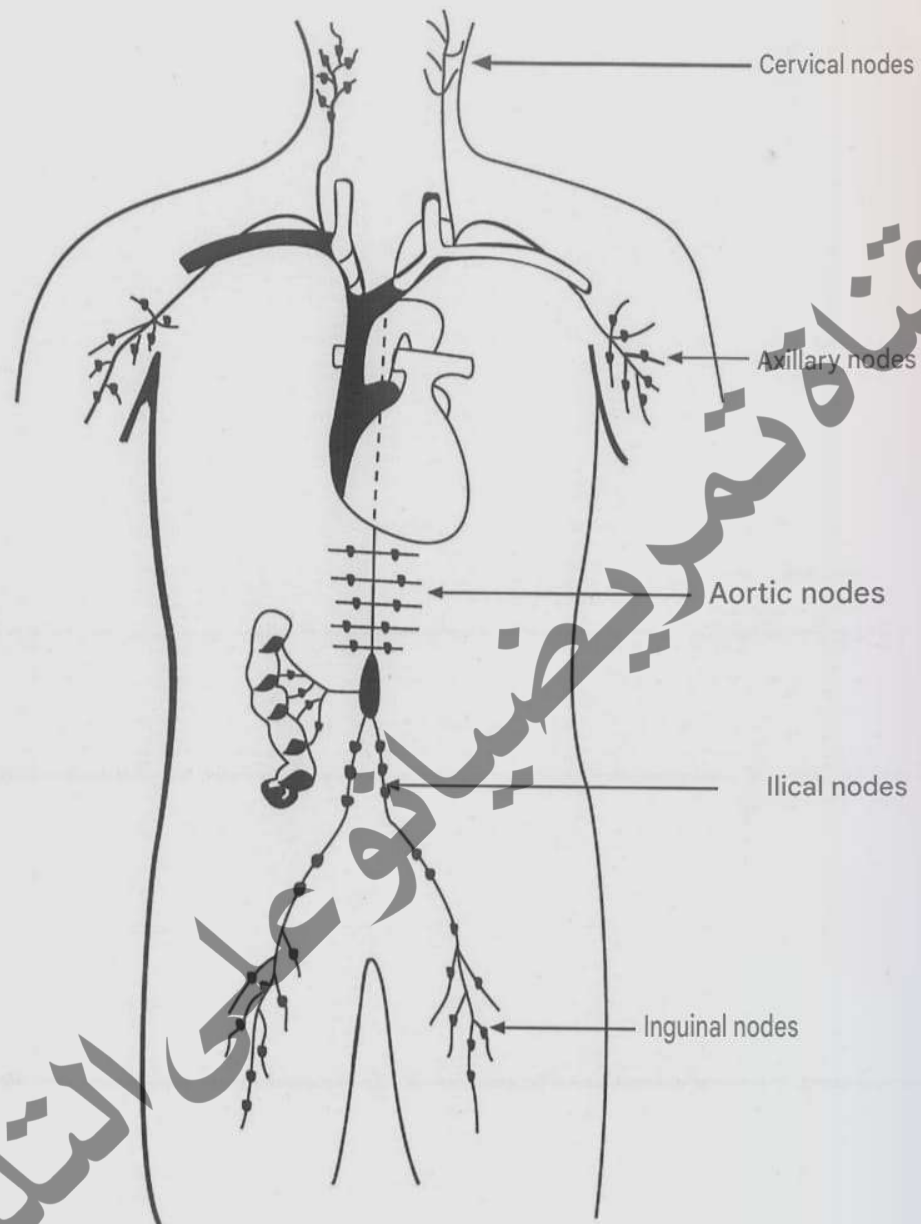
Very large in children (pressing on the heart) then gradually atrophies until it is difficult to see in adults.

Produces a special type of lymphocyte.

Internal surface of the spleen



Main Lymph Node Groups



The most important groups of lymph nodes

REVIEW QUESTIONS CHAPTER 4

1. Answer the following questions:

- What are the functions of lymph nodes?
- Name the main locations of lymph node groups and the body parts each group is responsible for.
- What is the location of the spleen and what is its function?

2. Complete the following sentences:

- The lymphatic system includes and
- The lymph nodes responsible for the breast are, while the inguinal lymph nodes are responsible for the external genital organs.
- The spleen has two surfaces: an external surface which contacts and, and an internal surface which contacts,, and

3. Choose the correct answer from the following:

a) The spleen destroys:

- ☐ White blood cells.
- ☐ Red blood cells.
- ☐ Blood platelets.
- ☐ Lymph nodes.

b) The inguinal lymph nodes are responsible for:

- ☐ The breast.
- ☐ The upper limb.
- ☐ The neck.
- ☐ The lower limb.

c) The thymus gland belongs to:

- ☐ The lymphatic system.
 - ☐ The digestive system.
 - ☐ The respiratory system.
 - ☐ The nervous system.
-

CHAPTER FIVE

The Respiratory System

Composition of the Respiratory System:

The Nose: Separated from the mouth by the palate (called the roof of the mouth). It functions to purify and warm inspired air in addition to the sense of smell.

The Pharynx:

The pharynx consists of 3 parts:

1. **Nasopharynx:**
Located behind the nose and contains:
2. **Opening of the Eustachian tube:**
A canal connecting the middle ear to the nasopharynx.
It has two functions:
 1. Drain secretions from the middle ear to the pharynx.
 2. Equalize air pressure inside and outside the eardrum.
3. **Pharyngeal tonsil (Adenoid):**
Located in the roof. Its inflammation leads to adenoid formation.
4. **Oropharynx:**
Located behind the mouth and contains the palatine tonsils on its sides.
5. **Laryngopharynx:**
Located behind the larynx.

The Larynx:

The larynx is composed of several cartilages, the largest being the thyroid cartilage, which is called the Adam's apple because it is more prominent in males.

- Above the thyroid cartilage is the epiglottis.
- Below the thyroid cartilage is the cricoid cartilage.
- Inside the thyroid cartilage are the vocal cords.
- Outside the thyroid cartilage is the thyroid gland.

The Trachea:

Its length is 10 cm and it consists of 15-20 C-shaped cartilaginous rings completed posteriorly by involuntary muscles. These muscles contract in bronchial asthma, leading to breathing difficulty.

The Bronchi:

The right and left bronchi differ in shape as follows:

Comparison between the right and left bronchi

Aspect	Right Bronchus	Left Bronchus
Lumen	Wider	Narrower
Length	Shorter	Longer
Alignment	In line with the trachea	Not in line with the trachea

Consequently, any foreign body entering the trachea tends to go into the right bronchus.

Each bronchus divides into bronchioles that extend inside the lungs.

At the end of the bronchioles are the alveoli, where gas exchange occurs.

The Lungs:

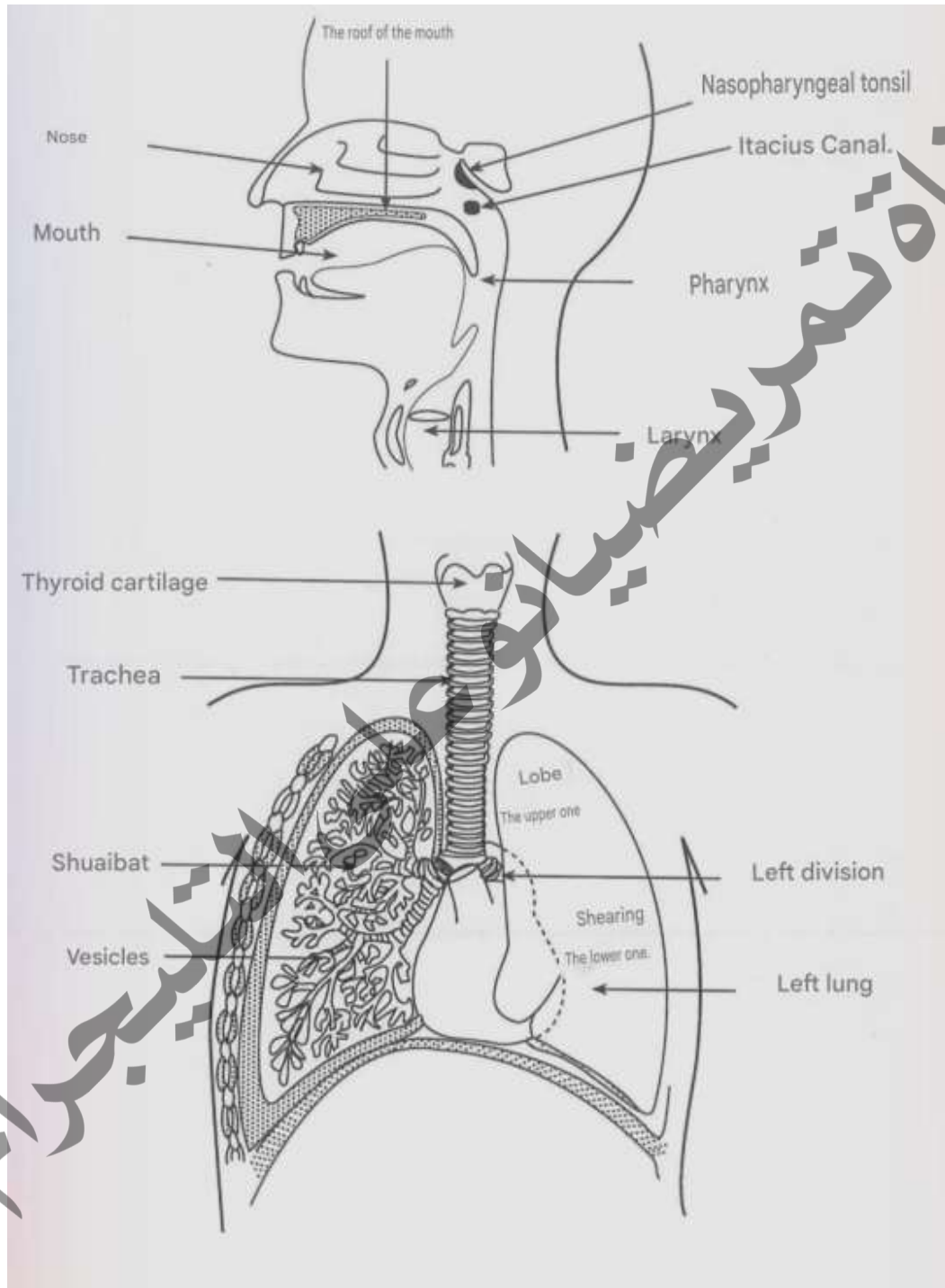
They are inside the thoracic cage.

The heart is surrounded by the pleural space, which consists of two thin layers with a potential space between them. Fluid can accumulate in this space in cases of pleural effusion, pressing on the lung and causing shortness of breath.

The following table compares the right and left lungs:

Aspect	Right Lung	Left Lung
Length	Short	Long
Width	Wide	Narrow
Weight	Heavy	Light
Fissures	Two (oblique & horizontal)	One (oblique)
Lobes	Three (Upper, Middle, Lower)	Two (Upper, Lower)
Cardiac Notch	Not present	Present (can be used to inject adrenaline directly into the heart muscle in cardiac arrest)
Base	Deep (due to liver pressure)	Not deep
Segments	10 segments total	8 segments total

The Respiratory System



REVIEW QUESTIONS CHAPTER 5

1. Answer the following questions:

- The pharynx consists of 3 parts. Mention what you know about them?
- Compare the right and left lungs.
- How is the larynx composed?

2. Complete the following sentences:

- The Eustachian tube carries and and has two benefits: and
- The thyroid cartilage is called the Adam's apple because and inside it are
- The right bronchus is characterized by being,, and
- The is surrounded by a membrane called in the cavity between the lungs.

3. Choose the correct answer from the following:

a) Located above the thyroid cartilage:

- ☐ The trachea.
- ☐ The epiglottis.
- ☐ The thyroid gland.
- ☐ The esophagus.

b) The right lung is characterized by:

- ☐ Being short.
- ☐ Having a cardiac notch.
- ☐ Having 8 small segments.
- ☐ Having only two lobes.

c) Number of lobes in the left lung:

- ☐ One.
 - ☐ Two.
 - ☐ Three.
 - ☐ Four.
-

CHAPTER SIX

The Digestive System

Consists of the following parts:

Mouth:

Contains the tongue, teeth, and openings of salivary glands (the parotid gland opens opposite the upper 2nd molar, while the other glands open in the floor of the mouth).

Teeth:

There are two types:

1. **Deciduous (Milk) Teeth:** 20 teeth total, 10 in each jaw. They start appearing at 6-8 months and are complete by age 2 years.
2. **Permanent Teeth:** 32 teeth total, 16 in each jaw. They start appearing at 6-7 years and are complete by age 12 years, except for the wisdom tooth which appears after 18 years.

Pharynx:

Mentioned previously with the respiratory system.

The Esophagus:

A muscular tube, 25 cm long. It connects the pharynx to the stomach.

It extends behind the trachea, then behind the heart, then pierces the diaphragm to connect to the stomach.

It has 3 narrow areas called esophageal constrictions.

Studying these constrictions is important because they are encountered when inserting a stomach tube for gastric lavage, feeding, or suctioning gastric secretions.

Constriction	Cause of Constriction	Distance from Teeth
First constriction	Cricoid cartilage	7 inches
Second constriction	Left bronchus	16 inches
Third constriction	Diaphragm	17 inches

The Stomach:

Consists of three parts: fundus - body - pylorus.

It has two surfaces: anterior and posterior.

And two ends: cardiac (connects to esophagus) - and pyloric (connects to the duodenum).

And two curvatures: the lesser curvature upwards and the greater curvature downwards.

The Small Intestine:

Includes:

1. **Duodenum:** Length 25 cm, shaped like the letter C. The common bile duct opens into it, carrying bile from the liver.
2. **Jejunum.**
3. **Ileum.**

Comparison between Jejunum and Ileum:

Aspect	Jejunum	Ileum
Length	Shorter (2.5 meters)	Longer (3.5 meters)
Width	Wider	Narrower
Contents	Often empty	Contains food

Parts of the Large Intestine:

1. Appendix.
2. Ascending colon.
3. Transverse colon.
4. Descending colon.
5. Rectum.

The appendix is located in the right iliac fossa and is known for causing acute inflammation without apparent cause or warning. Its treatment is immediate surgical removal before it ruptures

The following table compares the small and large intestine:

Feature	Small Intestine	Large Intestine
Length	6 meters	1.5 meters (150 cm)
Appendices Epiploicae	Not present	Present
Surface	Smooth	Puckered (Haustrated)
Teniae Coli	Not present	Present (3 bands)
Parts	3 parts: Duodenum Jejunum	8 parts: Appendix – Cecum - Ascending colon - Hepatic flexure Transverse colon - Splenic flexure - Descending colon

	Ileum	Sigmoid colon (Pelvic colon) – Rectum - Anal canal
--	-------	--

The Liver

The liver is the largest and heaviest gland in the body.

It is located in the right upper quadrant of the abdominal cavity, directly below the diaphragm.

It is wedge-shaped: its base is towards the right and its apex is towards the left, directly under the heart apex.

The liver consists of:

- **Lobes:** Two large lobes (right and left) and two small lobes.
- **Surfaces:** Superior - Inferior - Anterior - Posterior.

Blood vessels connected to the liver:

- **Enter from the inferior surface:**
 - **Portal Vein** (carries absorbed food from the intestines to the liver).
 - **Hepatic Artery** (carries O₂ necessary for liver cells).
- **Exit from the superior surface:** Two **Hepatic Veins** (carry CO₂ and empty into the Inferior Vena Cava).

The Biliary System

Biliary Ducts:

Bile secreted by the liver exits via the right and left hepatic ducts. These two ducts unite to form the common hepatic duct, which joins the cystic duct from the gallbladder to form the common bile duct.

Common Bile Duct:

Its length is 10 cm. It opens into the duodenum through a common opening with the pancreatic duct.

Gallbladder:

Located below the liver.

Its function: Storage and concentration of bile.

The Pancreas

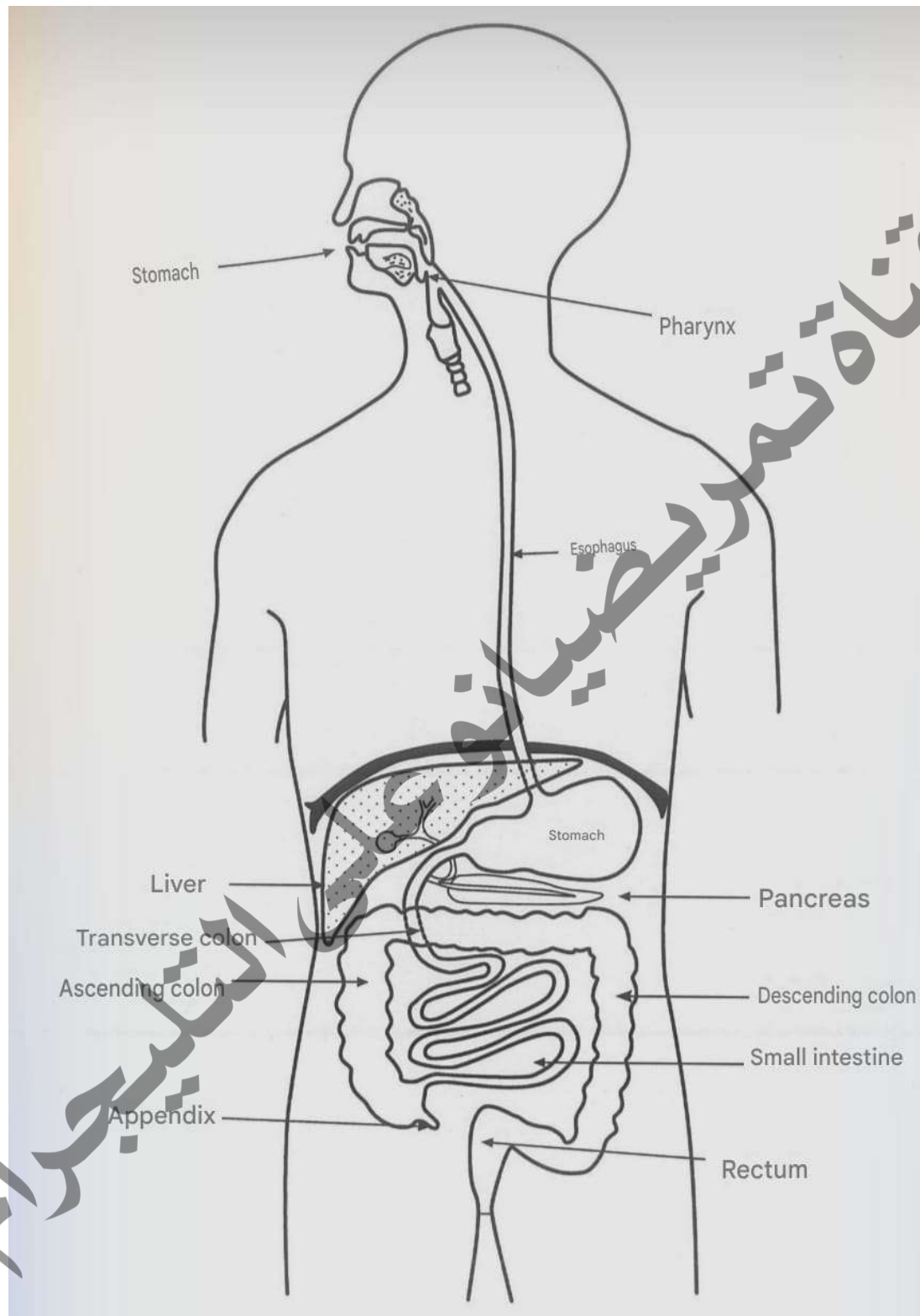
A gland consisting of 4 parts: head – neck – body – tail.

Location:

The head is surrounded by the duodenum, while the rest of the parts lie behind the stomach.

Its functions:

1. Secretes into the blood: the hormone Insulin.



REVIEW QUESTIONS CHAPTER 6

1. Answer the following questions:

- Discuss the parts, location, and function of the pancreas?
- Mention what you know about the three constrictions of the esophagus and state their importance.
- Compare the jejunum and ileum.
- Compare the small and large intestine.
- Name the lobes and surfaces of the liver. What blood vessels are connected to it?

2. Complete the following sentences:

- Deciduous teeth number ... in each jaw and begin to appear at age ... and are complete by age While permanent teeth number ... in each jaw and begin to appear at age and are complete by age
- The esophagus is a muscular tube cm long. It connects and

3. Choose the correct answer from the following:

a) The first constriction of the esophagus is caused by:

- () Thyroid cartilage.
- () Cricoid cartilage.
- () Left bronchus.
- () Diaphragm.

b) The common bile duct opens into:

- () The stomach.
- () The duodenum.
- () The jejunum.
- () The ileum.

CHAPTER SEVEN

The Urinary System

Composition:

The Kidneys:

Location:

- They are located in the abdomen, directly below the diaphragm.
- On either side of the vertebral column.
- They lie behind all other viscera, attached to the back muscles internally.
- The right kidney is slightly lower than the left due to pressure from the liver above it.

Parts of the Kidney:

1. Cortex: The outer part.
2. Medulla: The inner part.
3. Renal Pelvis: The beginning of the ureter.

A large renal artery from the aorta enters each kidney. A large renal vein exits each kidney and empties into the inferior vena cava.

The Ureters:

Length: 25-30 cm. They connect the kidney to the urinary bladder.

Urine moves through the ureter by contractions of its muscles. If there is a stricture or obstruction due to a stone, these contractions increase, leading to renal colic.

The Urinary Bladder:

Pyramidal in shape.

The ureters enter it from behind, and the urethra exits from its inferior aspect.

Location: In the pelvis when empty. But when full, it ascends into the abdomen.

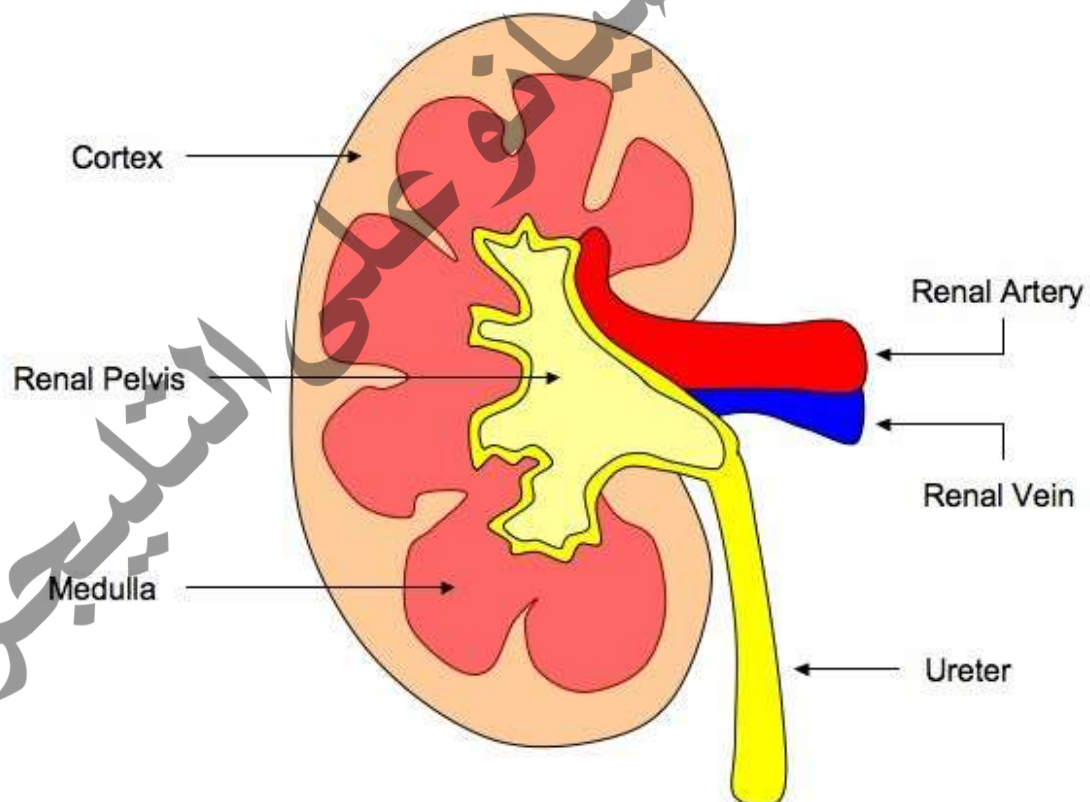
The Urethra:

The passage through which urine exits from the urinary bladder to the outside. It is much longer in males than in females.

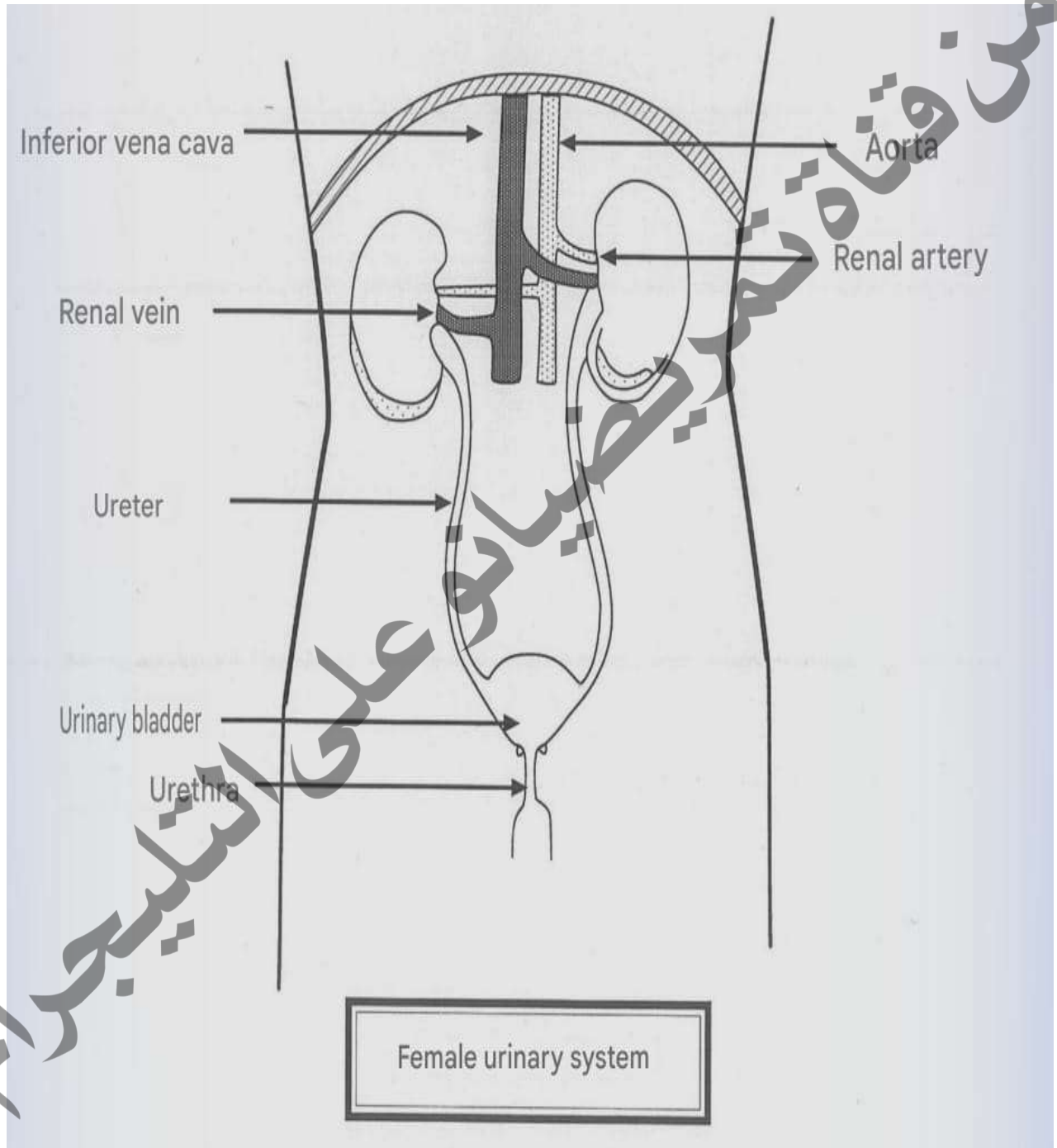
Comparison of the urethra in males and females:

Aspect	In Males	In Females
Length	20 cm	4 cm
Parts	1. Prostatic part (inside the prostate, length 4 cm). 2. Membranous part (between two membranes, length 1.5-2 cm). 3. Penile part (inside the penis, length 15 cm).	Has no parts
Function	1. Urine excretion 2. Semen excretion	Urine excretion only

Longitudinal section showing parts of the kidney



The Urinary System in the Female



REVIEW QUESTIONS CHAPTER 7

1. Answer the following questions:

- What is the location of the kidneys and what are the parts of a kidney?
- Describe the shape and location of the urinary bladder.
- Compare the urethra in males and females.

2. Complete the following sentences:

- The right kidney lies on the side and is than the left kidney due to
- The renal artery arises from while the renal vein empties into
- The ureter is long and connects and

3. Choose the correct answer from the following:

a) The length of the male urethra is:

- () 4 cm.
- () 10 cm.
- () 15 cm.
- () 20 cm.

b) The kidney is located:

- () In front of the vertebral column.
- () Beside the vertebral column.
- () In the pelvis.
- () In front of the stomach.

c) The longest part of the male urethra is the:

- () Prostatic part.
 - () Membranous part.
 - () Penile part.
 - () Pelvic part.
-

CHAPTER EIGHT

The Male Reproductive System

Internal Male Reproductive Organs:

Testes:

- **Location:** In the scrotum.
- **Function:**
 - Produce spermatozoa (sperm).
 - Secrete the male hormone (testosterone).

Epididymis:

A very narrow tube, 6 meters long, coiled upon itself to form a body 5 cm long. Sperm mature as they pass through it.

- **Location:** Inside the scrotum – behind the testis.
- **Its parts:** Head, body, and tail.

Vas Deferens (Ductus Deferens):

Length 45 cm, it connects the epididymis to the ejaculatory duct.

Seminal Vesicles:

Store sperm and secrete part of the seminal fluid to nourish the sperm inside them.

- **Location:** Behind the urinary bladder.

Ejaculatory Duct:

Considered an extension of the vas deferens inside the prostate. It pierces the prostate gland to open into the prostatic part of the urethra.

Male Accessory Glands:

They secrete seminal fluid in which sperm move and are nourished.

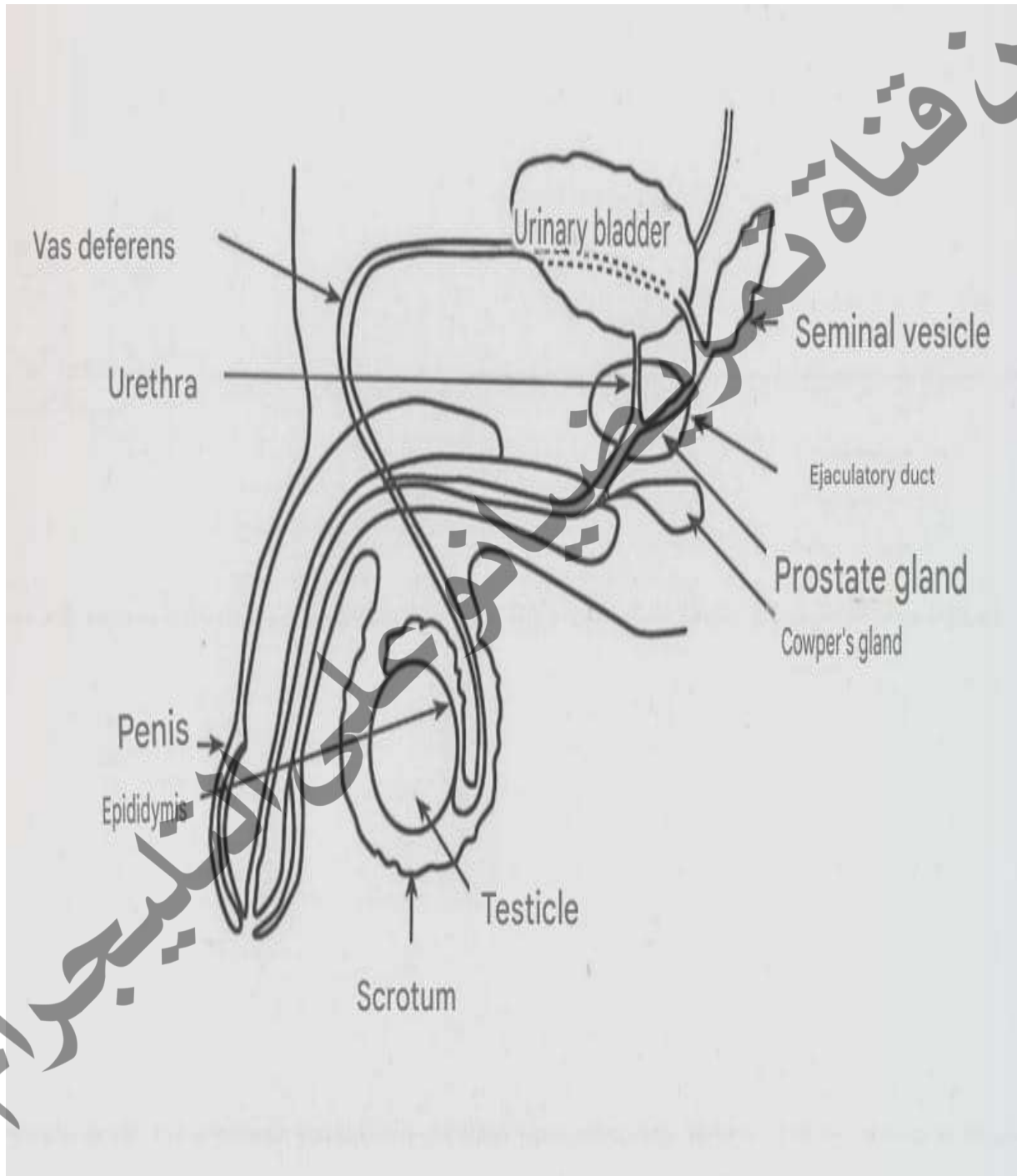
They include: the prostate gland, Cowper's glands (bulbourethral glands), and the seminal vesicles.

1. **Prostate Gland:** Located below the bladder. (It is the largest and most important one).
2. **Cowper's Gland (Bulbourethral Gland):** Located below the prostate.
3. **Seminal Vesicle:** Located behind the urinary bladder (also acts as a reservoir for sperm).

External Male Reproductive Organs:

Consist of the penis and scrotum.

The Male Reproductive System



REVIEW QUESTIONS CHAPTER 8

1. Answer the following questions:

- Mention the location of the testis and its function?
- What is the length of the epididymis, where is it located, and what is its function?
- What are the male accessory glands and where are they located?

2. Complete the following sentences:

- The vas deferens is long and connects and
- Semen consists of from the, which swim in a fluid from,, and
- The scrotum contains and

3. Choose the correct answer from the following:

a) The prostate gland is located:

- ☐ Behind the testis.
- ☐ Behind the urinary bladder.
- ☐ Inside the scrotum.
- ☐ Below the urinary bladder.

b) The epididymis is a narrow tube with a length of:

- ☐ 6 meters.
- ☐ 5 cm.
- ☐ 45 cm.
- ☐ 1 inch.

c) The epididymis is located:

- ☐ Behind the bladder.
 - ☐ Behind the testis.
 - ☐ Below the bladder.
 - ☐ Below the prostate.
-

CHAPTER NINE

The Female Reproductive System

Internal Female Reproductive Organs:

Ovaries:

Location: At the outer end of the fallopian tube.

Function:

1. Produce ova (eggs).
2. Secrete the female hormone (estrogen) and the pregnancy hormone (progesterone).

Fallopian Tubes:

Length: 10 cm.

Their outer end: embraces the ovary.

Their inner end: connects to the uterus at its outer angle.

Their functions:

1. The ovum passes through them from the ovary towards the uterus.
2. Sperm pass through them from the uterus towards the ovary.
3. Fertilization (union of ovum and sperm) occurs inside them near their outer end.

Uterus:

A pear-shaped muscular organ. Located in the pelvis. The urinary bladder is in front of it and the rectum is behind it.

Its parts: Fundus – Body – Cervix.

Dimensions: 3 x 2 x 1 inches.

Its functions:

1. Menstruation (the monthly shedding of part of the uterine lining with a little blood).
2. Pregnancy occurs inside it when the embryo implants in its lining, which then stabilizes and menstruation stops.

Vagina:

A canal 6-7 cm long connected to the cervix and opening externally into the vulva.

It has a membrane near its external opening called the hymen, which contains one or more holes for the passage of menstrual blood.

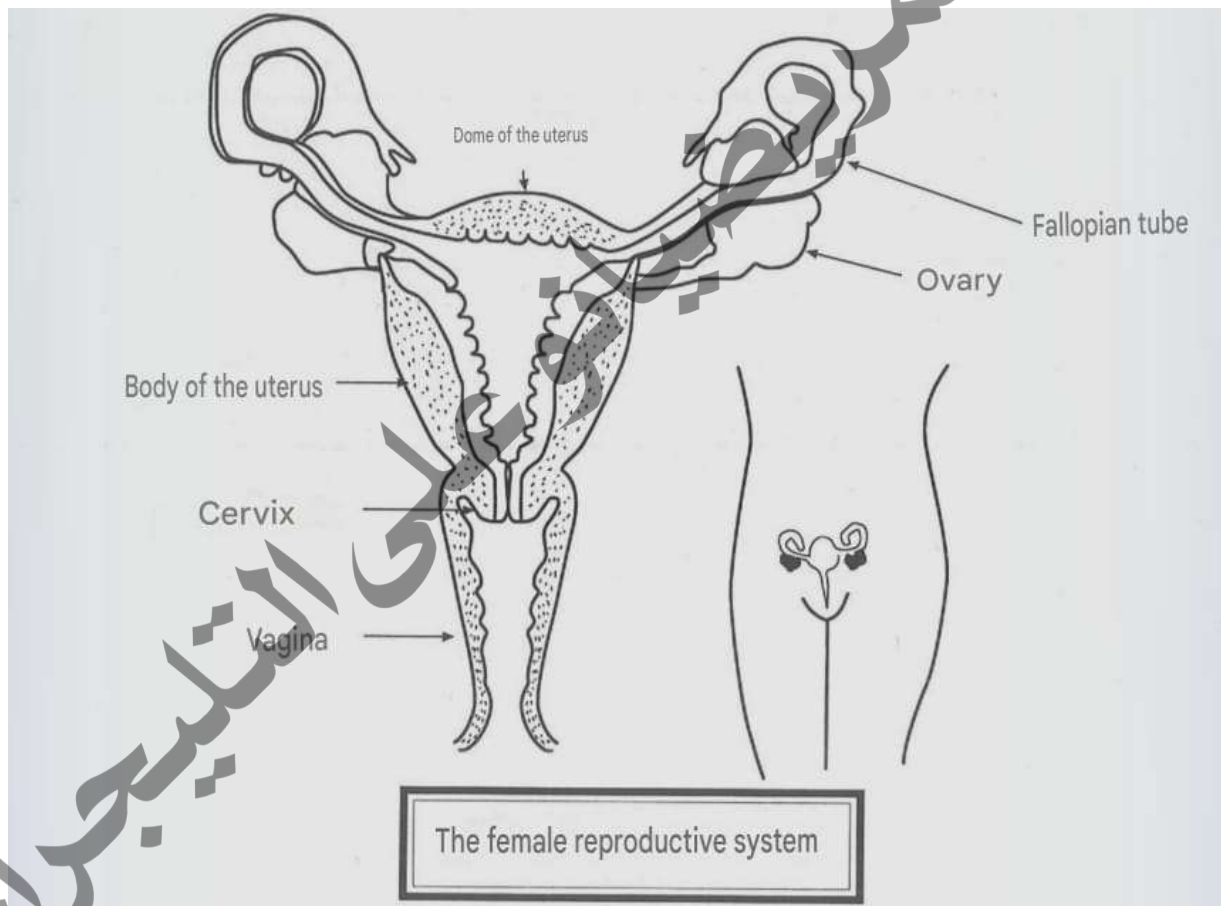
Its wall is stretchable to allow the passage of the child during birth.

Females have Bartholin's glands (or vulvovaginal glands) on either side of the lower part of the vagina. Their ducts open into the vulva on either side of the vaginal opening.

External Female Reproductive Organs:

- Labia majora.
 - Labia minora: located inside the labia majora.
The space between them is called the vulva.
 - The vulva contains:
 - The vaginal opening (in the middle).
 - The openings of Bartholin's glands (on either side of the vaginal opening).
 - The urethral opening (in front of the vaginal opening).
 - The clitoris: a small organ resembling the male penis (in front of the urethral opening).
-

The Female Reproductive System



REVIEW QUESTIONS CHAPTER 9

1. Answer the following questions:

- Mention the location and function of the ovary.
- What is the length of the fallopian tube and what is its function?
- Mention the dimensions of the uterus and its function?
- What openings are present in the vulva?

2. Complete the following sentences:

- The uterus is located in the pelvis between in front and behind.
- The uterus is long and consists of three parts:,, and
- The ovary is located at the end of the and secretes the hormones and

3. Choose the correct answer from the following:

a) The highest part of the uterus is called:

- ☐ The fundus.
- ☐ The body.
- ☐ The cervix.
- ☐ The vulva.

b) Fertilization occurs in:

- ☐ The vagina.
- ☐ The uterus.
- ☐ The ovary.
- ☐ The fallopian tube.

c) The length of the fallopian tube is:

- ☐ 4 cm.
 - ☐ 10 cm.
 - ☐ 4 inches.
 - ☐ 25 cm.
-

CHAPTER TEN

The Glands

Types of Glands in the Body:

Glands are divided according to their method of secretion into:

1. **Exocrine Glands (Duct Glands):**

Secrete through ducts either to the outside or into a body cavity. Examples:

1. The three salivary glands secrete saliva into the mouth.
2. The prostate gland secretes seminal fluid.
3. The lacrimal glands secrete tears.
4. Skin glands: such as sweat glands, sebaceous glands, and ceruminous glands in the external ear canal.
5. The breast, which secretes milk.

2. **Endocrine Glands (Ductless Glands):**

Secrete hormones directly into the blood. Examples:

1. **Pituitary Gland:** Located below the brain and secretes:

- a. Growth hormone.
- b. Hormone that controls urine output (ADH).
- c. Hormone that controls milk secretion from the breast (Prolactin).
- d. Hormone that helps contract the uterus during childbirth (Oxytocin).
- e. Hormones that control other endocrine glands (Tropic hormones).

2. **Thyroid Gland:** Located in the neck on either side of the larynx. Secretes thyroxine.

3. **Parathyroid Glands:** There are two on each side, located behind the thyroid gland.

4. **Adrenal Glands (Suprarenal):** Located above the kidney. Secrete the hormones adrenaline and cortisol.

3. **Mixed Glands:**

Combine features of both exocrine and endocrine glands. Examples:

1. **Pancreas:** Located behind the stomach. Secretes: the hormone insulin (endocrine) and pancreatic juice (exocrine).
 2. **Testis:** Secretes the male hormone (endocrine) and sperm (exocrine).
 3. **Ovary:** Secretes the female hormone and pregnancy hormone (endocrine) and ova (exocrine).
-

CHAPTER ELEVEN

The Breast

The breast is one of the glands that open onto the skin surface.

The breast is underdeveloped in children of both sexes. At puberty, it remains underdeveloped in males while it grows and develops in females. Further development occurs in the breast during the first pregnancy in preparation for milk secretion for the newborn.

Breast Boundaries:

- Medially: The edge of the sternum.
- Laterally: A vertical line passing through the middle of the axilla.
- Superiorly: The 2nd rib.
- Inferiorly: The 6th rib.

The breast rests posteriorly on the strong chest muscles.

Internal Structure of the Breast:

The breast consists of three types of tissue:

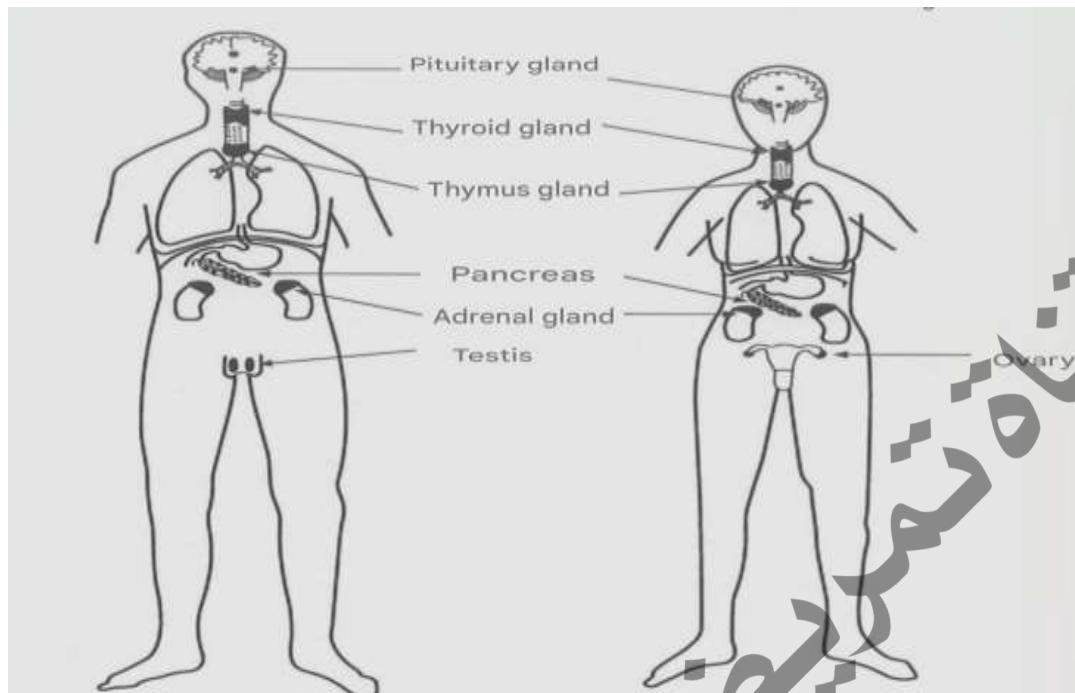
1. **Glandular tissue:** Inside the breast are 15-20 lobes. Each lobe consists of several lobules connected to a lactiferous duct that opens at the nipple tip with an independent opening. This means the tip of the nipple has 15-20 small openings, each releasing the secretion of its corresponding lobe.
2. **Fibrous tissue:** Consists of fibrous bands extending between the lobes.
3. **Adipose tissue:** Fills the rest of the breast.

External Shape Features of the Breast:

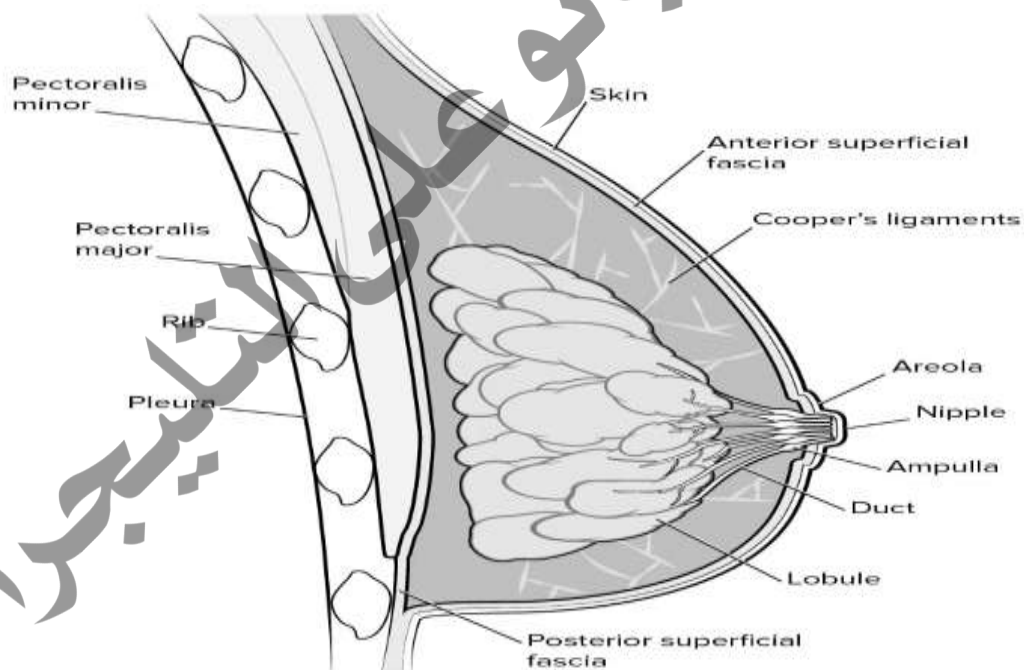
1. **Nipple:** Its tip contains 15-20 small openings, each connected to the lactiferous duct of one of the 15-20 lobes. The nipple contains involuntary muscles that allow it to become erect when the baby starts breastfeeding. The nipple is surrounded by a pink areola which turns brown during the first pregnancy.
2. **Axillary Tail of the Breast (Tail of Spence):** An extension from the upper outer part of the breast into the axilla. Some may mistake it for an enlarged lymph node.

The lymphatic vessels of the breast drain into the axillary lymph nodes, and thus these lymph nodes swell in cases of infections or malignant tumors of the breast.

Major Endocrine and Mixed Glands



Longitudinal Section of the Breast



REVIEW QUESTIONS CHAPTER 10 & 11

1. Answer the following questions:

- What is the difference between exocrine glands and endocrine glands?
- Where is the pituitary gland located and what hormones does it secrete?
- What are the mixed glands in the body?
- Mention the boundaries of the area where the breast is located.

2. Complete the following sentences:

- Exocrine glands secrete their products through while endocrine glands secrete directly, and their secretions are called
- A part of the breast extends into the axilla and is called the
- The breast consists of three types of tissue:,,
- The skin contains many exocrine glands such as,

3. Choose the correct answer from the following:

a) The thyroid gland is located:

- () On either side of the mandible.
- () On either side of the larynx.
- () Above the kidney.
- () Behind the stomach.

b) The lymphatic vessels of the breast drain into:

- () The cervical lymph nodes.
- () The aortic lymph nodes.
- () The axillary lymph nodes.
- () The inguinal lymph nodes.